

Lecture Notes on Modern QFT

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Lecture notes toward the quantum field theory textbook of ~ 2046 : the book that will exist once the developments of 2004–2026 — generalized and non-invertible symmetries, the bootstrap renaissance, monotonicity theorems, the algebraic/entanglement revival, the lattice–continuum program — are digested into pedagogy the way Wilson’s revolution was digested into the textbooks of the 1990s. The notes open with an annotated table of contents (twelve parts, fifty-five chapters): each chapter comes with a paragraph explaining why it exists and pointers into the current literature (Seiberg, Witten, Shao, Córdova, Schäfer-Nameki, Rychkov, Simmons-Duffin, Komargodski, Gaiotto, Rastelli, Yin, Polyakov’s heirs all). Chapters are then being written one at a time, starting with Part I. The cited papers are collected in `assets/`, indexed by arXiv number; see `assets/README.md`.

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The premise

Every generation's textbook teaches quantum field theory as that generation understood it. The canon of the 1990s–2000s (Peskin–Schroeder, Weinberg) presents QFT as a procedure: start from a classical Lagrangian, quantize, renormalize, expand around free fields. This is how one *computes* in a QFT that happens to be weakly coupled somewhere — but twenty years of work have made it untenable as a statement of what a QFT *is*. There are interacting theories with no known Lagrangian (the $(2,0)$ theory, most of class $\mathcal{S}$); single theories with several inequivalent Lagrangians (the 2+1d duality web); symmetries invisible in any Lagrangian (non-invertible defects); and lattice models whose continuum limits violate the lore the Lagrangian paradigm is built on (fractons). Meanwhile the most precise determination of the 3d Ising exponents comes not from any Lagrangian manipulation but from unitarity and crossing alone.

The premise of these notes: imagine the standard graduate text of c. 2046, written after this reorganization is complete. What follows is its table of contents — twelve parts, fifty-five chapters — with annotations explaining what each chapter contains, why it sits where it sits, and where to learn the material *today*. It is, deliberately, a reading map of the 2012–2026 formal-QFT literature disguised as front matter. For the prehistory and spirit of this point of view, see Seiberg's essay [Thoughts About Quantum Field Theory](#) and Rychkov's recent history of the bootstrap, [2509.02779](#). A note on references: hyperlinked arXiv numbers like [2308.00747](#) refer to papers downloaded into `assets/`; plain typewriter numbers like 2307.07547 are further reading left on the arXiv.

Ten organizing principles

The book is organized around ten lessons, each of which would have read as heterodox in 2000 and reads as obvious in 2046:

1. **A theory is its observables, not its Lagrangian.** Lagrangians, lattices, bootstrap data, holographic duals are *presentations*; the theory is the abstract structure they present. Some theories have many presentations (dualities), some have none we know.
2. **Defects are part of the definition.** The operator content of a QFT includes extended operators of every codimension — lines, surfaces, interfaces, boundaries — and theories with identical local correlators can differ in their extended sectors.
3. **The arena is theory space.** The renormalization group is its dynamics, conformal field theories are its landmarks, and effective field theory is the statement that physics is locally well-approximated by coordinates around a fixed point.
4. **Symmetry = topological operators.** This single shift unifies ordinary symmetries, higher-form symmetries, higher groups, and non-invertible symmetries, and makes symmetry a property of the theory rather than of its presentation. Symmetries form (higher) categories; gauging is an operation on theory space.
5. **Anomalies are the most robust observables in physics.** 't Hooft anomalies are exactly RG-invariant, are classified by topological field theories in one higher dimension, and constrain dynamics at any coupling.
6. **Consistency alone computes.** Unitarity, crossing, and symmetry carve out the possible: the conformal bootstrap solves strongly-coupled CFTs to high precision, and dispersive positivity bounds delineate which EFTs can ever be UV-completed.

7. **Duality is the rule, not the exception.** The same point of theory space typically admits several weakly-coupled descriptions valid in different regimes, related by gauging operations.
8. **Quantization is not fundamental.** Since there are QFTs that are not quantizations of anything classical, any acceptable definition of the subject must not start from a classical theory.
9. **The continuum is subtle.** Lattice symmetries *emanate* into internal continuum symmetries; Lieb–Schultz–Mattis theorems are lattice ’t Hooft anomalies; fracton models have continuum limits with UV/IR mixing; ϕ^4 in $d = 4$ is rigorously trivial. “Take a continuum limit” is a theorem in some cases, a conjecture in most, and false in interesting ones.
10. **QFT is quantum information organized by geometry.** Local algebras are Type III, entanglement proves monotonicity theorems and energy conditions, and gravity — QFT’s largest customer via holography — repays the debt by reorganizing its algebras.

How the parts fit together

Parts I–IV are the spine: the arena (I), the exact structures on it (II), its landmarks (III), and its dynamics (IV). Parts V–VIII are dynamics in the flesh: gauge theories (V), the exactly solvable corners that anchor everything (VI), duality (VII), and the lattice–continuum dictionary where the definition of the subject is stress-tested (VIII). Parts IX–XI are interfaces with quantum information and gravity. Part XII is the frontier. A reader of 2026 wanting one path: $I \rightarrow II \rightarrow III.1\text{--}III.2 \rightarrow IV.1 \rightarrow V \rightarrow VIII$, with VI and IX read in parallel.

Three one-semester courses are embedded in the book: *symmetry-first QFT* (Parts I, II, V, VIII — the Seiberg school); *the bootstrap* (Parts I, III, IV, X — the Polyakov program executed); *fields, information, and gravity* (Parts I, IX, XI). Three lighthouse theories recur in every part and should be solved completely, once, early: the 2d Ising model with its Majorana chain (symmetry, duality, the lattice), the 3d Ising CFT (bootstrap, defects, fuzzy sphere), and Maxwell/Yang–Mills in 4d (higher-form symmetry, θ , lines). When these notes graduate from table of contents to text, the writing should start where today’s textbooks are thinnest: Parts II and VIII exist only as lecture notes and research papers.

Part I. The Arena: Quantum Field Theory Without Lagrangians

What is being defined before any method of computing it: observables, states, and the space of theories.

I.1 What is a quantum field theory?

The question, taken seriously. Why the textbook answer of 2000 (“quantize a Lagrangian”) fails to cover the known examples; the inventory of things a QFT assigns (correlators, Hilbert spaces, defects, partition functions); QFT as the universal description of many degrees of freedom with locality. A first look at the picture that organizes the whole book: theories as points, RG as flow, fixed points as landmarks.

Sections: the standard model of textbooks and its discontents; what we must assign and to what; locality as the one non-negotiable; theory space as the picture to keep.

Key reading: Dedushenko [2203.08053](#) (survey of every known definition-attempt); Rychkov [2509.02779](#) (how we got here); Seiberg, *Thoughts About QFT* (IAS essay).

I.2 Observables I: local operators

The local operator content as primary data. Correlation functions, the operator product expansion as an exact (convergent, in CFT) structure rather than an asymptotic trick; Euclidean vs. Lorentzian, Wightman functions vs. Schwinger functions, reflection positivity; the stress tensor and currents as the operators every local QFT shares.

Sections: correlators and their analytic structure; OPE; reconstruction theorems; the stress tensor; conserved currents and their products.

Key reading: Rychkov [1601.05000](#) (chs. 1–2); Kontsevich–Segal [2105.10161](#) (§2, for which Euclidean backgrounds are allowed).

I.3 Observables II: defects and extended operators

The lesson of 2014–2026: the definition of a theory includes its extended operators. Order vs. disorder operators; Wilson and ’t Hooft lines; surface operators; interfaces and boundaries; defects as probes (one-point functions of bulk operators, defect-localized degrees of freedom) and as charged objects for the symmetries of Part II.

Sections: codimension as organizing principle; genuine vs. non-genuine (attached) operators; defect Hilbert spaces; line operators as phase diagnostics; boundaries as the richest defects.

Key reading: Northe [2411.03381](#) (pedagogical BCFT/DCFT); Aharony–Seiberg–Tachikawa [1305.0318](#) (why the line spectrum is part of the definition).

I.4 States, geometry, and cutting-and-gluing

The Hilbert space is assigned to a spatial slice, and the path integral is the gluing rule: the functorial (Atiyah–Segal) skeleton of QFT, stated physically. Radial quantization and the state–operator map as the CFT incarnation. Which complex metrics are allowed (Kontsevich–Segal–Witten) — the modern answer to “what does Wick rotation mean?”

Sections: Hilbert spaces from slices; cutting and gluing; the cylinder vs. the sphere; state–operator correspondence; allowable complex backgrounds; thermal states and KMS.

Key reading: Kontsevich–Segal [2105.10161](#); Dedushenko [2203.08053](#) (§§on functorial QFT).

I.5 Presentations, not definitions

One theory, many descriptions. The same point in theory space presented as: a Lagrangian flowed from a free fixed point; a lattice model at criticality; abstract CFT data $\{\Delta_i, c_{ijk}\}$; an integrable S-matrix; a holographic bulk. Worked example threaded through the book: the 3d Ising CFT as ϕ^4 , as the lattice model, as a bootstrap island, as the fuzzy-sphere spectrum.

Sections: Lagrangians as charts; lattice presentations; CFT data; S-matrices; holographic presentations; when presentations disagree about what is elementary.

Key reading: Poland–Rychkov–Vichi [1805.04405](#) (§1, the data-centric view); Zhu et al. [2210.13482](#) (the same CFT from a Landau level).

I.6 Theory space and the renormalization group

Wilson’s picture promoted to the book’s geography. RG flow on the space of theories; fixed points, relevant/irrelevant/marginal directions as the tangent space; universality as the statement that flows converge; effective field theory as local coordinates; conformal manifolds; lines of fixed points; the questions the rest of the book answers (what is exactly transported along flows? — Part II; what are the fixed points? — Part III; which flows are allowed? — Part IV).

Sections: Wilsonian RG without a cutoff fetish; scheme-independence and what is physical; EFT; universality classes; conformal manifolds and exactly marginal deformations.

Key reading: Rychkov [1601.05000](#) (ch. 1); Casini–Huerta [2201.13310](#) (§1 for the entropic restatement, used in Part IX).

Part II. Symmetry

The single largest reorganization since 2000: symmetry as topology, anomalies as exact data. This part is the technology the rest of the book runs on.

II.1 Symmetries are topological operators

Noether’s theorem, run backwards: a symmetry *is* a topological codimension-one operator $U_g(\Sigma)$, defined with no reference to a Lagrangian. Ward identities from deforming surfaces; action on local and extended operators; spontaneous breaking restated operatorially; coupling to background gauge fields as the universal bookkeeping device.

Sections: from currents to topological surfaces; Ward identities topologically; symmetry defects and twisted sectors; SSB without order parameters; background fields.

Key reading: Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#) (§1–2); Shao [2308.00747](#) (ch. 1); Iqbal [2407.20815](#) (gentlest entry).

II.2 Gauging and ’t Hooft anomalies

Gauging as summing over backgrounds; ’t Hooft anomalies as the obstructions to doing so, and anomaly *matching* as the resulting exact RG invariant — the sharpest nonperturbative tool in the subject. Anomaly inflow: anomalies of a d -dimensional theory are classified by invertible/SPT phases

in $d + 1$ dimensions. Perturbative (chiral) and global (parity, time-reversal, discrete) anomalies in one framework.

Sections: gauging finite and continuous symmetries; 't Hooft's argument; inflow and the bulk-boundary picture; SPT phases and the cobordism classification (sketch); discrete anomalies; examples from QCD to the honeycomb lattice.

Key reading: Shao [2308.00747](#); Schäfer-Nameki [2305.18296](#); Córdova–Dumitrescu–Intriligator–Shao [2205.09545](#) (panorama with full bibliography).

Further: Witten–Yonekura (1909.08775) for η -invariants and the deepest version of inflow.

II.3 Higher-form symmetries and the global structure of gauge theories

Symmetries whose charged objects are extended: one-form symmetries acting on lines, etc. Maxwell theory's electric and magnetic one-form symmetries; the center symmetry of Yang–Mills, with confinement reborn as unbroken one-form symmetry (the Landau paradigm regained for gauge theories); $SU(N)$ vs. $SU(N)/\mathbb{Z}_N$ and discrete θ -parameters — the global structure that correlators of local operators cannot see; two-groups and symmetry fusion across dimensions.

Sections: q -form symmetries; charged defects; selection rules for extended operators; the center of Yang–Mills; reading between the lines; 2-group symmetry.

Key reading: Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#); Aharony–Seiberg–Tachikawa [1305.0318](#); Iqbal [2407.20815](#) (applications incl. magnetohydrodynamics).

II.4 Non-invertible symmetries

Topological operators without inverses: fusion replaces composition, categories replace groups. The Kramers–Wannier defect of the 2d Ising model as the founding example; topological defect lines in 2d CFT; duality defects in 3+1d gauge theories at self-dual couplings; the chiral symmetry of massless QED resurrected as a non-invertible symmetry (the ABJ anomaly tamed); selection rules and the meaning of “symmetry” once unitarity of the action is abandoned.

Sections: fusion categories in physics; Kramers–Wannier; Verlinde lines; half-gauging and duality defects in 4d; non-invertible chiral symmetry; what non-invertible symmetries forbid.

Key reading: Shao [2308.00747](#) (the standard course); Choi–Córdova–Hsin–Lam–Shao [2111.01139](#) (4d duality defects).

Further: Kaidi–Ohmori–Zheng (2111.01141); Córdova–Ohmori (2205.06243) and Choi et al. (2205.05086) for the Standard Model; Bhardwaj et al. (2307.07547) and Brennan–Hong (2306.00912) as alternative lecture notes.

II.5 The Symmetry Topological Field Theory

The organizing gadget of the 2020s: the symmetry content of a d -dimensional theory encoded in a $(d+1)$ -dimensional topological theory with two boundaries — one topological (the symmetry and its realization), one dynamical (the physics). Gauging, discrete torsion, and duality webs as changes of topological boundary condition; anomalies as obstructions visible in the bulk; the kinematics/dynamics separation made into a picture.

Sections: the sandwich; topological boundary conditions; gauging as boundary change; anomalies in the bulk; reading phase diagrams from the SymTFT.

Key reading: Schäfer-Nameki [2305.18296](#) (§SymTFT); Kapustin-Seiberg [1401.0740](#) (the seed: coupling a QFT to a TQFT).

Further: Freed-Moore-Teleman ([2209.07471](#)) for the mathematical formulation.

II.6 Anomalies in parameter space

Anomalies generalized from symmetries to families: topological terms for *background* scalars — coupling constants — and the exact statements they yield. The θ -angle as a circle-valued coupling with an anomaly in its periodicity; consequences: level crossings that cannot be avoided, walls that must carry degrees of freedom, constraints on axion physics.

Sections: families of theories as one theory with background scalars; anomalies valued in one higher dimension of parameter space; θ -periodicity; axions; defects from winding couplings.

Key reading: Córdova-Freed-Lam-Seiberg [1905.09315](#).

II.7 The generalized Landau paradigm

Phases of matter re-founded on the full symmetry category. Gapped phases as (symmetry-enriched) TQFTs; spontaneous breaking of higher-form and non-invertible symmetries; topological order as ordinary SSB of one-form symmetry; anomalies forbidding trivially gapped phases (symmetry-protected criticality); the condensed-matter dictionary.

Sections: what replaces the order parameter; gapped phases with categorical symmetry; deconfined criticality (preview); symmetric mass generation; the dictionary with condensed matter.

Key reading: McGreevy [2204.03045](#); Córdova-Dumitrescu-Intriligator-Shao [2205.09545](#).

Part III. Fixed Points: Conformal Field Theory

The landmarks of theory space, studied as theories-in-themselves with no microscopic crutch. Polyakov's 1974 program, finally executed.

III.1 Conformal kinematics in $d \geq 3$

The conformal algebra and its unitary representations; primaries and descendants; radial quantization and the state-operator map (the cleanest argument for the OPE's convergence); unitarity bounds; correlators fixed by symmetry up to functions of cross-ratios; embedding space.

Sections: conformal algebra and representations; radial quantization; unitarity bounds; two- and three-point functions; conformal blocks; the convergent OPE.

Key reading: Rychkov [1601.05000](#); Simmons-Duffin [1602.07982](#) (the standard modern courses).

III.2 The conformal bootstrap

Crossing symmetry plus unitarity as a computational engine. The numerical method (positivity, linear/semidefinite programming) and its conceptual meaning: rigorous exclusion bounds on theory space, no Lagrangian anywhere. The 3d Ising island as the flagship: critical exponents to six digits from consistency alone. The frontier: navigator methods, larger systems of correlators, the stress tensor.

Sections: crossing equations; the positivity trick; islands and kinks; 3d Ising and $O(N)$; rigor of numerics; current frontiers.

Key reading: Simmons-Duffin [1602.07982](#); Poland–Rychkov–Vichi [1805.04405](#); Rychkov–Su [2311.15844](#) (state of the art); Rychkov [2509.02779](#) (the story).

Further: El-Showk et al. (1203.6064), the paper that started the modern era.

III.3 The analytic bootstrap

Crossing solved in expansions: large spin, lightcone limits, and the Lorentzian inversion formula — the theorem that CFT data is analytic in spin, with the Regge trajectory as the fundamental object. Dispersive sum rules; what analyticity buys that numerics cannot, and conversely.

Sections: lightcone bootstrap and large spin; the inversion formula; Regge behavior in CFT; dispersive sum rules; interplay with numerics.

Key reading: Caron-Huot [1703.00278](#); Hartman et al. [2202.11012](#) (Snowmass survey of the whole program).

Further: Caron-Huot–Mazáč–Rastelli–Simmons-Duffin (2008.04931) for dispersive sum rules.

III.4 Two dimensions

The exactly solvable floor of the subject, taught as a modern topic rather than folklore: Virasoro symmetry, minimal models, modular invariance and the Cardy formula; the modular bootstrap; Liouville theory; topological defect lines as the original non-invertible symmetries (closing the loop with Part II); the irrational frontier.

Sections: Virasoro representation theory; minimal models; modular invariance and modular bootstrap; Liouville; defect lines in 2d; what 2d does and does not teach us about $d > 2$.

Key reading: Yin, *Aspects of 2d CFTs* (TASI, in `assets/`); Kusiuki [2412.18307](#) (a genuinely modern course).

Further: Ribault (1406.4290) for the systematic treatment.

III.5 Boundaries, defects, and interfaces in CFT

Conformal defects as CFTs’ richest substructure: BCFT and the Cardy condition; defect operator expansions and bulk-to-defect data; line defects as quantum mechanics coupled to a CFT (the Wilson line as a renormalization problem); the defect bootstrap; defect RG flows and the g -theorem (proved entropically in Part IX).

Sections: BCFT; Cardy states; defect CFT data and crossing; magnetic lines in the 3d Ising model; defect RG.

Key reading: Northe [2411.03381](#); Billò–Gonçalves–Lauria–Meineri (1601.02883).

III.6 New ways to solve a CFT

The methods invented after the textbooks were written. Fuzzy-sphere regularization: 3d CFTs from a single Landau level, with the state–operator map realized in a quantum simulator’s Hilbert space. The large-charge expansion: strongly-coupled CFTs made semiclassical sector by sector. Hamiltonian truncation. What “solving” means when there was never a coupling to expand in.

Sections: fuzzy sphere: setup, spectrum, defects; large charge as controlled semiclassics; Hamiltonian truncation; cross-validation among methods and with the bootstrap.

Key reading: Zhu et al. [2210.13482](#); Álvarez-Gaumé–Orlando–Reffert [2008.03308](#).

Further: Hellerman et al. (1505.01537), the large-charge original.

Part IV. Flows: The Renormalization Group as Dynamics

What can be said exactly about the flow between landmarks: monotonicity, matching, and the strange solvable deformations.

IV.1 Monotonicity theorems

Irreversibility of the RG made precise: Zamolodchikov’s c -theorem in 2d; the a -theorem in 4d via the dilaton effective action (anomaly matching weaponized); the F -theorem in 3d, where no anomaly exists and quantum information takes over; defect analogues (g - and b -theorems). What these theorems say about theory space (no cycles; a partial order) and what they do not.

Sections: c ; a via dilatons; F and sphere partition functions; entropic unification (preview of Part IX); defect monotonicity; uses: ruling out IR scenarios.

Key reading: Komargodski–Schwimmer [1107.3987](#); Casini–Huerta [2201.13310](#); Klebanov–Pufu–Safdi (1105.4598).

IV.2 Constraining the deep infrared

The modern toolkit applied to the question “what happens at long distances?”: anomaly matching forcing massless modes or symmetry breaking or TQFTs; the possible IR endpoints classified (gapped/TQFT, CFT, SSB with Goldstones, free); emergent symmetries in the IR and their anomalies; worked examples where the answer is known and where it is genuinely open.

Sections: the list of possible IR fates; matching as a sieve; emergent symmetry; examples: QCD chiral symmetry breaking; adjoint QCD₂; open cases.

Key reading: Gaiotto–Kapustin–Komargodski–Seiberg [1703.00501](#); Córdova–Dumitrescu–Intriligator–Shao [2205.09545](#) (§applications).

Further: Komargodski–Ohmori–Roumpedakis–Seifnashri (2008.07567) for adjoint QCD₂ done with full modern machinery.

IV.3 The edge of criticality

Fixed points are born, collide, and annihilate: merger of fixed points and the loss of conformality; complex CFTs and walking behavior; weakly first-order transitions masquerading as second-order; conformal windows of gauge theories. The RG as a dynamical system, with theory space’s bifurcations as physics.

Sections: fixed-point annihilation; complex CFTs; walking; conformal windows; dangerously irrelevant operators; diagnostics in practice.

Key reading: Rychkov–Su [2311.15844](#) (§on conformal windows); Gorbenko–Rychkov–Zan (complex CFTs; see references in the above).

IV.4 Solvable deformations and the integrable corner

Flows we can follow exactly. Integrable deformations of 2d theories; the $T\bar{T}$ deformation: irrelevant yet solvable, with a universal energy-level flow and a UV that is not a fixed point — the cleanest known counterexample to “irrelevant means uncontrollable,” and a glimpse of physics beyond Wilsonian QFT (resumed in Part XII).

Sections: integrable flows; $T\bar{T}$: definition, Burgers equation for spectra, interpretations; what replaces the UV fixed point; cousins ($J\bar{T}$ etc.).

Key reading: Smirnov–Zamolodchikov (1608.05499); Jiang [1904.13376](#) (pedagogical).

Part V. Gauge Dynamics

Gauge theory with the new tools: global structure, extended order parameters, anomalies at $\theta = \pi$. The questions of the 1970s, finally answerable in part.

V.1 What is a gauge theory?

Gauge “symmetry” as redundancy of presentation, not symmetry — and gauging as an operation that maps one global-symmetry-bearing theory to another. The choices hidden in “gauge it”: the global form of the group, discrete θ -angles, the spectrum of lines. Emergent gauge fields in condensed matter as proof that gauge structure is descriptive, not fundamental.

Sections: gauging as an operation (finite groups first); choices and discrete data; line spectra again; emergent gauge fields; why “gauge symmetry breaking” is a category error stated carefully.

Key reading: Kapustin–Seiberg [1401.0740](#); Aharony–Seiberg–Tachikawa [1305.0318](#).

V.2 Phases of gauge theories

The phase diagram problem organized by extended operators and one-form symmetry: Wilson loops, area vs. perimeter law, confinement as unbroken center symmetry; Higgs–confinement complementarity and where it fails; oblique phases; ’t Hooft’s classification revisited with modern eyes.

Sections: order parameters that work; one-form SSB = deconfinement; Higgs vs. confining; dyonic condensation and oblique phases; finite temperature.

Key reading: Gaiotto–Kapustin–Seiberg–Willett [1412.5148](#) (§4); McGreevy [2204.03045](#) (condensed-matter side).

V.3 Theta

The θ -angle as the laboratory for everything in Parts II and IV: Yang–Mills at $\theta = \pi$ and its mixed time-reversal/center anomaly — an exact statement about a famously intractable theory; domain walls forced to be nontrivial; large- N θ -dependence; the strong CP problem restated with anomalies in parameter space.

Sections: θ -dependence generalities; the $\theta = \pi$ anomaly; walls and interfaces; large N ; axions and Part II.6 revisited.

Key reading: Gaiotto–Kapustin–Komargodski–Seiberg [1703.00501](#); Córdova–Freed–Lam–Seiberg [1905.09315](#).

V.4 Low-dimensional laboratories

Where strong coupling is tame enough to dissect: the Schwinger model as the exactly solvable confining theory; adjoint QCD₂, where non-invertible symmetries control screening vs. confinement; $\mathbb{CP}^{N-1}$ and instantons; 3d Chern–Simons-matter; deconfined quantum criticality. Each laboratory keyed to which 4d lesson it teaches.

Sections: Schwinger model; adjoint QCD₂; sigma models with theta terms; CS-matter; deconfined criticality; the laboratory-to-4d dictionary.

Key reading: Komargodski–Ohmori–Roumpedakis–Seifnashri (2008.07567); Seiberg–Senthil–Wang–Witten 1606.01989 (3d).

V.5 The Standard Model through modern eyes

The SM as a case study in everything above: the global form of its gauge group and the line operators that distinguish the candidates; its higher-form and non-invertible symmetries ($B - L$, the axial $U(1)$); anomaly-based exact statements relevant to phenomenology; naturalness questions restated symmetry-first.

Sections: which SM? (global structures); magnetic lines and completeness; non-invertible symmetries of the SM; pions from anomaly matching; what is and is not explained.

Key reading: Córdova–Dumitrescu–Intriligator–Shao 2205.09545 (§SM); Choi et al. (2205.05086); Córdova–Ohmori (2205.06243).

Part VI. Exactly Solvable Corners: Supersymmetry and Friends

Solvable islands are not the point of the subject; they are its lighthouses — the places where proposed general principles get checked exactly.

VI.1 Holomorphy and Seiberg duality

$\mathcal{N} = 1$ theories as the minimal setting where strong-coupling questions have exact answers: holomorphy as the supersymmetric replacement for perturbation theory; moduli spaces of vacua; Seiberg duality as the prototype of IR duality — and the historical seed of the entire duality-centric worldview.

Sections: holomorphy arguments; exact superpotentials; moduli spaces; Seiberg duality and its checks; SUSY breaking in brief.

Key reading: Tachikawa 1312.2684 (companion volume); Intriligator–Seiberg lectures (classic, see references therein).

VI.2 Seiberg–Witten theory

The first complete solution of the IR of a strongly-coupled 4d gauge theory: the Coulomb branch as a fibered geometry, monodromies, monopole condensation as the mechanism of confinement made exact. Presented as the template for “solving” a QFT: identify protected data, find the geometry that encodes it.

Sections: the Coulomb branch; the SW curve; monopole condensation and confinement; breaking to $\mathcal{N} = 1$; the template abstracted.

Key reading: Seiberg–Witten hep-th/9407087; Tachikawa 1312.2684.

VI.3 Localization and exact partition functions

Path integrals computed exactly: supersymmetric localization on spheres, indices, and the precision-duality era they enabled. What localization computes (protected data), what it cannot, and how its outputs feed monotonicity theorems (F) and bootstrap baselines.

Sections: the localization argument; sphere partition functions; superconformal indices; precision tests of duality; limits of the method.

Key reading: Pestun (0712.2824); the review volume *Localization Techniques in QFT* (1608.02952).

VI.4 Theories without Lagrangians

The existence proof that broke the old definition: the 6d $(2, 0)$ theory; class $\mathcal{S}$ — 4d $\mathcal{N} = 2$ theories from punctured surfaces, with duality frames as pair-of-pants decompositions; Argyres–Douglas points. Why “the space of QFTs” is strictly larger than “the space of Lagrangians,” and how one works with a theory that has no path integral.

Sections: the $(2, 0)$ theory; compactification as theory factory; class $\mathcal{S}$ and duality made geometric; Argyres–Douglas; how to compute without a Lagrangian.

Key reading: Gaiotto [0904.2715](#); Tachikawa [1312.2684](#).

VI.5 Protected subsectors and integrability

Exact substructures inside non-integrable theories: the chiral algebra (2d VOA) inside every 4d $\mathcal{N} = 2$ SCFT — Rastelli school — feeding exact CFT data into the bootstrap; topological sectors; integrability of planar $\mathcal{N} = 4$ as the maximal solvable corner; 2d factorized S-matrices (used again in Part X).

Sections: the 4d/2d chiral-algebra map; protected operator algebras as bootstrap input; planar integrability in one chapter; factorized scattering.

Key reading: Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees [1312.5344](#).

Part VII. Duality

If a theory is not its presentation, different presentations of one theory must be expected. They are.

VII.1 What duality means

Duality as equality of all observables under a dictionary; exact dualities (Kramers–Wannier, T-duality, $\mathcal{N} = 4$ S-duality) vs. IR dualities (Seiberg, the 3d web); what must match (anomalies, symmetry categories, line spectra) and how proposed dualities are stress-tested with the Part II toolkit.

Sections: definitions and gradations of duality; the matching checklist; anomalies as duality invariants; historical exemplars.

Key reading: Seiberg–Senthil–Wang–Witten [1606.01989](#) (§1–2).

VII.2 The 2+1d duality web

Bosonization in three dimensions: particle–vortex duality old and new, fermion–boson dualities, and the web generated from a seed duality by gauging — with condensed-matter incarnations (quantum Hall transitions, surface states of topological insulators) as both motivation and evidence.

Sections: particle–vortex; the seed duality and the web; flux attachment; condensed-matter applications; status of the evidence.

Key reading: Seiberg–Senthil–Wang–Witten [1606.01989](#); Senthil–Son–Wang–Xu (review; see references therein).

VII.3 Duality from gauging

The mechanism beneath the webs: dualities generated by gauging discrete symmetries (orbifolding), with Kramers–Wannier as the original example; self-duality producing non-invertible symmetry (Part II.4 closed into a loop); the SymTFT as the bookkeeping that makes whole duality webs one picture.

Sections: gauging $\mathbb{Z}_2$ in 2d done three ways; orbifolds; self-duality $\Rightarrow$ duality defects; webs from the SymTFT sandwich.

Key reading: Kapustin–Seiberg [1401.0740](#); Choi–Córdova–Hsin–Lam–Shao [2111.01139](#); Shao [2308.00747](#) (§KW).

Part VIII. The Lattice and the Continuum

If QFTs are defined by continuum limits of lattice models, the dictionary between lattice and continuum had better be understood exactly. It is subtler than anyone assumed.

VIII.1 Continuum limits, reexamined

What “taking the continuum limit” asserts; universality as the (usually unproven) theorem behind it; which structures pass to the limit (symmetries? anomalies? Hilbert spaces?) and which arise only there. The program: treat the lattice not as a regulator but as a definition, and audit everything.

Sections: lattice models as definitions; universality audited; symmetries of the lattice vs. of the limit; emergent symmetry and its breaking by irrelevant operators.

Key reading: Cheng–Seiberg [2211.12543](#) (§1); Dedushenko [2203.08053](#) (§lattice).

VIII.2 Emanant symmetries

Seiberg–Shao: continuum internal symmetries that *emanate* from lattice translations — not emergent (violated at finite spacing) but exact consequences of lattice symmetry acting differently in the limit. The Majorana chain and Ising model worked completely; anomalies of the emanant symmetry predicted from the lattice; non-invertible translation symmetry at criticality.

Sections: emergent vs. emanant; the Majorana chain in detail; Ising and its KW operator from lattice translation; anomaly inheritance; consequences for numerics.

Key reading: Seiberg–Shao [2307.02534](#).

VIII.3 Lieb–Schultz–Mattis theorems as anomalies

LSM-type theorems — lattice systems forbidden from having trivially gapped ground states — understood as ’t Hooft anomaly matching between lattice translation and internal symmetry; Luttinger’s theorem in the same frame; the non-invertible refinement on tensor-product Hilbert spaces. Condensed matter and formal QFT collapsing into one subject.

Sections: LSM classically; the anomaly interpretation; filling constraints and Luttinger; non-invertible LSM; applications to spin chains and beyond.

Key reading: Cheng–Seiberg [2211.12543](#); Seiberg–Seifnashri–Shao [2401.12281](#).

VIII.4 Exotica: fractons and UV/IR mixing

The systems that break the continuum lore: fracton models with subsystem symmetries, restricted mobility, and ground-state degeneracies that depend on the lattice size — continuum limits in which discontinuous field configurations dominate and rotation invariance never returns. What, exactly, must be relaxed in the axioms; the modified Villain construction as the bridge.

Sections: the X-cube and friends; subsystem symmetry; UV/IR mixing; continuum descriptions with discontinuous fields; modified Villain lattices; what fractons teach about “QFT”.

Key reading: Seiberg–Shao (2003.10466); Gorantla–Lam–Seiberg–Shao (2103.01257); both summarized in Córdova et al. [2205.09545](#).

VIII.5 Rigor: constructive QFT and triviality

What mathematics actually knows: constructive successes in $d = 2, 3$; the Aizenman–Duminil-Copin theorem that ϕ_4^4 and 4d Ising have (marginally) trivial scaling limits — a rigorous statement about the Standard Model’s scalar sector; Yang–Mills existence as a Millennium problem; chiral fermions on the lattice as the deepest unsolved regulator problem.

Sections: what has been constructed; triviality in $d = 4$ and its meaning; the YM problem stated honestly; Nielsen–Ninomiya and chiral lattice fermions; rigorous bootstrap bounds as a new kind of rigor.

Key reading: Aizenman–Duminil-Copin [1912.07973](#); Dedushenko [2203.08053](#).

Part IX. Quantum Fields and Quantum Information

The Hilbert space of a QFT is not a big collection of qubits: its entanglement structure is the theory.

IX.1 Entanglement in quantum field theory

Entanglement entropy of regions: UV divergences and what is universal; replica trick; exact results in 2d and for free fields; mutual information and relative entropy as the cutoff-free quantities; entanglement as the modern order parameter (topological entanglement entropy, anomalies detected by entanglement).

Sections: regions and reduced states; the replica trick; universal terms; relative entropy and its monotonicity; entanglement diagnostics of phases.

Key reading: Casini–Huerta [2201.13310](#); Witten [1803.04993](#).

IX.2 The algebraic viewpoint

Why regions of a QFT have no density matrices: local algebras, Type III, and the failure of factorization — stated as physics, not pathology; Tomita–Takesaki modular theory and what modular flow means (Bisognano–Wichmann: it is a boost); the type classification as a physical classification of how systems entangle.

Sections: algebras instead of tensor factors; Types I/II/III with physical fingerprints; modular flow; Bisognano–Wichmann; half-sided modular inclusions (a glimpse).

Key reading: Witten [1803.04993](#); Sorce [2302.01958](#); Liu [2510.07017](#) (lectures, with the gravity payoff).

IX.3 Information-theoretic theorems

The harvest: entropic proofs of the c - and F -theorems (strong subadditivity doing the work anomalies cannot in odd d); the averaged null energy condition proved from causality and from modular theory; QNEC; Bekenstein-type bounds from relative entropy. Energy conditions — once assumptions — now theorems of quantum information.

Sections: entropic c and F ; ANEC two ways; QNEC; Bekenstein from relative entropy; what remains conjectural.

Key reading: Casini–Huerta [2201.13310](#) (incl. [hep-th/0405111](#)); Hartman–Kundu–Tajdini ([1610.05308](#)); Faulkner et al. ([1605.08072](#)).

IX.4 Coupling to gravity: the crossed product

What happens to the algebras when gravity is turned on, however weakly: Witten’s question “why does QFT in curved spacetime make sense?”; the crossed-product construction rendering Type III into Type II, giving finite (renormalized) entropies; observers as part of the algebra. The bridge into Part XI.

Sections: QFT in curved spacetime as an algebra problem; the crossed product; finite generalized entropies; observers; de Sitter remarks.

Key reading: Witten [2112.11614](#); Liu [2510.07017](#).

Part X. Scattering, the UV, and the Space of Consistent Theories

The S-matrix program, reborn as a bootstrap: carve out the space of consistent theories from unitarity, analyticity, and crossing — in flat space this time.

X.1 The S-matrix bootstrap

The 1960s program with 2020s technology: nonperturbative two-to-two amplitudes constrained by unitarity + analyticity + crossing, explored numerically (primal and dual); pion physics as a target; QFT in AdS as the regulator connecting the S-matrix bootstrap to the conformal bootstrap of Part III.

Sections: axioms for amplitudes; numerical primal/dual methods; the pion bootstrap; QFT in AdS; relation to the conformal bootstrap.

Key reading: Kruczenski–Penedones–van Rees [2203.02421](#); Paulos et al. ([1607.06109](#)).

X.2 Positivity and the UV–IR connection

Which low-energy EFTs admit any UV completion at all: dispersion relations, positivity of forward amplitudes, the classic IR obstruction (Adams et al.), and the modern sharpening into two-sided bounds and the EFT-hedron geometry of Wilson coefficients — the “swampland program” of pure QFT, with experimental teeth.

Sections: dispersion relations; forward positivity; two-sided bounds; the EFT-hedron; causality constraints; applications from gravity to particle physics.

Key reading: Adams–Arkani-Hamed–Dubovsky–Nicolis–Rattazzi [hep-th/0602178](#); de Rham et al. (2203.06805); Arkani-Hamed–Huang–Huang (2012.15849).

X.3 Toward flat-space holography

The frontier chapter: soft theorems reorganized as symmetries of the S-matrix; celestial and Carrollian proposals for a holographic description of flat space; what is established, what is aspirational, and what the S-matrix bootstrap says either way.

Sections: soft theorems as Ward identities; celestial sphere kinematics; status report, written skeptically.

Key reading: Kruczenski–Penedones–van Rees [2203.02421](#) (§outlook); Strominger’s lectures (see references therein).

Part XI. Large N , Holography, and Gravity

Large N is a classical limit no Lagrangian advertises, and holography is its destiny: QFT as the definition of quantum gravity.

XI.1 Large N as a classical limit

Factorization of correlators at large N as the emergence of a classical theory — whose phase space is nothing like the classical limit $\hbar \rightarrow 0$; vector vs. matrix models; planarity; the master field; $1/N$ as the loop-counting parameter of an emergent weakly-coupled description (the bulk, in waiting).

Sections: vector models solved; planar counting; factorization = classicality; large- N phase transitions; what $1/N$ organizes.

Key reading: Penedones [1608.04948](#) (§large N); ’t Hooft’s original logic as retold there.

XI.2 Holography from the CFT side

AdS/CFT presented as a statement *within* QFT: which CFTs have local bulk duals (large N + large gap); Heemskerk–Penedones–Polchinski–Sully; the holographic dictionary derived from CFT axioms; Mellin amplitudes; bootstrap bounds becoming statements about gravity (graviton couplings, higher-derivative corrections).

Sections: the HPPS criterion; the dictionary from the CFT side; Mellin space; bulk locality from crossing; gravity constrained by CFT consistency.

Key reading: Heemskerk–Penedones–Polchinski–Sully [0907.0151](#); Penedones [1608.04948](#).

Further: the holographic correlators program (Rastelli–Zhou and successors; see Penedones’s lectures and references therein).

XI.3 Quantum field theory as quantum gravity

The strongest claim in the book: in AdS, quantum gravity is *defined* by a QFT, making everything in Parts I–IX a statement about quantum gravity. What gravity has paid back: the chaos bound, energy conditions, the algebraic reorganization of Part IX.4; black-hole information as a theorem about unitary QFT.

Sections: AdS/CFT as definition; information-paradox dividends; the chaos bound as a QFT theorem; gravity’s feedback into pure QFT; limits of the definition (dS, cosmology) as segue to Part XII.

Key reading: Penedones [1608.04948](#); Maldacena–Shenker–Stanford (1503.01409).

Part XII. Epilogue: The Next Twenty Years

What the 2046 book admits it does not know — the chapters that will be rewritten by 2066.

XII.1 The definitional frontier

The axiomatizations compared honestly: Wightman/Osterwalder–Schrader; Haag–Kastler nets; Atiyah–Segal functorial QFT and Kontsevich–Segal allowability; factorization algebras. Each captures a fragment; none captures class $\mathcal{S}$, large N , *and* fractons at once. The definition of QFT as an open research problem, not an embarrassment.

Sections: the contenders; what each proves; the counterexample stress-tests; is one definition even desirable?

Key reading: Dedushenko [2203.08053](#); Kontsevich–Segal [2105.10161](#).

XII.2 Computational frontiers

Where new answers will come from: quantum simulation of QFTs (the Jordan–Lee–Preskill program) and the regulator questions it forces (Part VIII reappears as engineering); tensor networks; bootstrap numerics at scale; machine-learned exclusion plots. Computation as experiment for formal QFT.

Sections: quantum simulation of scattering; lattice regularizations for hardware; tensor networks and criticality; the numerical bootstrap’s next decade.

Key reading: Jordan–Lee–Preskill (1111.3633); Rychkov–Su [2311.15844](#) (§outlook).

XII.3 Beyond quantum field theory

The boundary of the subject mapped from inside: little string theories (local energy density without local fields); $T\bar{T}$ -deformed theories as UV-complete non-QFTs; QFT in de Sitter and the cosmological bootstrap — correlators at the end of inflation as the next “S-matrix”; gravity as the deformation that finally leaves theory space.

Sections: symptoms of non-QFT UV behavior; little strings; $T\bar{T}$ revisited as a beyond-QFT laboratory; dS and cosmological observables; what replaces theory space.

Key reading: Jiang [1904.13376](#); Arkani-Hamed et al. (1811.00024); Seiberg’s talks on what lies beyond QFT.

XII.4 Open problems

A concrete list, stated so that progress is checkable: (1) Yang–Mills existence and mass gap — now with anomaly-matching constraints any solution must obey; (2) a nonperturbative definition of the full Standard Model, i.e. chiral fermions on the lattice; (3) classify 3d and 4d CFTs, or even all theories with c (or a) below a given value; (4) prove the a -theorem’s gradient-flow refinement, and monotonicity in $d = 6$; (5) solve the 3d Ising model analytically; (6) determine whether the space of QFTs is connected; (7) complete the classification of symmetry categories and their gapped/gapless realizations in $d \geq 4$; (8) glueball masses from the S-matrix bootstrap; (9) a sharp nonperturbative formulation of observables in de Sitter; (10) a definition of QFT wide enough for class $\mathcal{S}$ and fractons simultaneously — or a proof that none exists.

Key reading: every entry above, and the next twenty years of `arXiv:hep-th`.

The library

All papers below are in `assets/`, with filenames beginning with the arXiv number; `assets/README.md` maps them to the parts above. Lecture notes and reviews are marked ●; foundational research papers ○.

Framing and history

- Dedushenko, *Snowmass: The Quest to Define QFT*, [2203.08053](#)
- Kontsevich–Segal, *Wick Rotation and the Positivity of Energy in QFT*, [2105.10161](#)
- Aizenman–Duminil-Copin, *Marginal Triviality of ϕ_4^4* , [1912.07973](#)
- Rychkov, *Conformal Bootstrap: from Polyakov to Our Times*, [2509.02779](#)

Symmetry (Part II)

- Gaiotto–Kapustin–Seiberg–Willett, *Generalized Global Symmetries*, [1412.5148](#)
- Córdova–Dumitrescu–Intriligator–Shao, *Snowmass: Generalized Symmetries*, [2205.09545](#)
- Shao, *TASI Lectures on Non-Invertible Symmetries*, [2308.00747](#)
- Schäfer–Nameki, *ICTP Lectures on (Non-)Invertible Generalized Symmetries*, [2305.18296](#)
- Iqbal, *Jena Lectures on Generalized Global Symmetries*, [2407.20815](#)
- McGreevy, *Generalized Symmetries in Condensed Matter*, [2204.03045](#)
- Aharony–Seiberg–Tachikawa, *Reading Between the Lines of 4d Gauge Theories*, [1305.0318](#)
- Kapustin–Seiberg, *Coupling a QFT to a TQFT and Duality*, [1401.0740](#)
- Choi–Córdova–Hsin–Lam–Shao, *Non-Invertible Duality Defects in 3+1d*, [2111.01139](#)
- Córdova–Freed–Lam–Seiberg, *Anomalies in the Space of Coupling Constants I*, [1905.09315](#)

Lattice ↔ continuum (Part VIII)

- Seiberg–Shao, *Majorana Chain and Ising Model: Emanant Symmetries*, [2307.02534](#)
- Cheng–Seiberg, *LSM, Luttinger, and 't Hooft*, [2211.12543](#)
- Seiberg–Seifnashri–Shao, *Non-Invertible Symmetries and LSM-Type Constraints*, [2401.12281](#)

CFT and bootstrap (Part III)

- Rychkov, *EPFL Lectures on CFT in $D \geq 3$* , [1601.05000](#)
- Simmons-Duffin, *TASI Lectures on the Conformal Bootstrap*, [1602.07982](#)
- Poland–Rychkov–Vichi, *The Conformal Bootstrap* (RMP), [1805.04405](#)

- Rychkov–Su, *New Developments in the Numerical Bootstrap*, [2311.15844](#)
- Hartman et al., *Snowmass: The Analytic Conformal Bootstrap*, [2202.11012](#)
- Caron-Huot, *Analyticity in Spin in Conformal Theories*, [1703.00278](#)
- Northe, *Maynooth Lectures on CFT, BCFT and DCFT*, [2411.03381](#)
- Zhu et al., *Uncovering Conformal Symmetry: the Fuzzy Sphere*, [2210.13482](#)
- Kusuki, *Modern Approach to 2D Conformal Field Theory*, [2412.18307](#)
- Yin, *Aspects of Two-Dimensional CFTs* (PoS TASI2017)

RG flows (Part IV)

- Komargodski–Schwimmer, *On RG Flows in Four Dimensions*, [1107.3987](#)
- Casini–Huerta, *Lectures on Entanglement in QFT*, [2201.13310](#)
- Jiang, *Pedagogical Review of Solvable Irrelevant Deformations*, [1904.13376](#)

Gauge dynamics and duality (Parts V, VII)

- Gaiotto–Kapustin–Komargodski–Seiberg, *Theta, Time Reversal, and Temperature*, [1703.00501](#)
- Seiberg–Witten, *Monopole Condensation and Confinement*, [hep-th/9407087](#)
- Seiberg–Senthil–Wang–Witten, *A Duality Web in 2+1d*, [1606.01989](#)

Exact corners (Part VI)

- Tachikawa, *$\mathcal{N} = 2$ Supersymmetric Dynamics for Pedestrians*, [1312.2684](#)
- Gaiotto, *$\mathcal{N} = 2$ Dualities*, [0904.2715](#)
- Beem et al., *Infinite Chiral Symmetry in Four Dimensions*, [1312.5344](#)

Quantum information (Part IX)

- Witten, *Notes on Some Entanglement Properties of QFT*, [1803.04993](#)
- Sorce, *Notes on the Type Classification of von Neumann Algebras*, [2302.01958](#)
- Witten, *Why Does QFT in Curved Spacetime Make Sense?*, [2112.11614](#)
- Liu, *Lectures on Entanglement, von Neumann Algebras, and Spacetime*, [2510.07017](#)

Scattering and the UV (Part X)

- Kruczenski–Penedones–van Rees, *Snowmass: S-matrix Bootstrap*, [2203.02421](#)
- Adams et al., *Causality, Analyticity and an IR Obstruction*, [hep-th/0602178](#)

Large N and holography (Part XI)

- Penedones, *TASI Lectures on AdS/CFT*, [1608.04948](#)
- Heemskerk–Penedones–Polchinski–Sully, *Holography from CFT*, [0907.0151](#)

Methods (Parts III.6, IV.4, and beyond)

- Álvarez-Gaumé–Orlando–Reffert, *The Large Quantum Number Expansion*, [2008.03308](#)
- Aniceto–Başar–Schiappa, *A Primer on Resurgent Transseries*, [1802.10441](#)

Part I

The Arena: Quantum Field Theory Without Lagrangians

I.1 What is a quantum field theory?

There is a question that every course on quantum field theory postpones, and that these notes take as their subject: what, exactly, *is* a quantum field theory? Not how to compute in one — you know how to do that. You can renormalize a loop integral, extract a cross section, run a coupling. The question is what the object is that all this machinery computes *about*.

For quantum mechanics the analogous question has an answer that fits on a postcard. A quantum system is a Hilbert space $\mathcal{H}$, states are rays in it, observables are self-adjoint operators, time evolution is a unitary e^{-iHt} . Every textbook states these axioms in chapter one; nothing discovered since 1930 has dislodged them. Now try to recall the corresponding list from your quantum field theory course. There isn't one. What you were given instead was a *recipe* — pick fields, write a Lagrangian, quantize, renormalize — together with the implicit promise that the recipe defines the subject. The first purpose of this chapter is to show, by exhibiting concrete and well-established physics, that it does not: there are quantum field theories the recipe cannot produce, single theories it produces in several inequivalent ways, and recipe-outputs that turn out not to exist. The second purpose is constructive: to assemble the list of things a quantum field theory actually provides — its observables — and to draw the picture that organizes everything else in these notes, in which theories are points in a space, the renormalization group is a flow on that space, and conformal field theories are its fixed points.

Question I.1.1 — What must we specify to specify a quantum field theory? And to what mathematical and geometric objects does a quantum field theory, once specified, assign physical data?

A warning about the spirit of what follows. Nothing in this chapter is a complaint about the practical success of the standard framework, which is the most quantitatively successful structure in the history of science and which we will use constantly. The claim is about logic, not utility: the standard framework is a description of how to *present* certain quantum field theories and expand them around free fields, and the subject has outgrown the identification of theories with their presentations. Making that distinction precise is worth a whole chapter, because almost every modern development — generalized symmetry, the bootstrap, duality, the lattice–continuum program — begins from it.

I.1.1 The standard paradigm

Let us first state the inherited answer honestly, because it deserves it. The textbook definition of a quantum field theory, circa 2000, is a procedure:

1. Choose fields: a collection of functions on spacetime transforming in representations of the Poincaré group and of some internal (possibly gauge) group.

2. Write the most general local Lagrangian density $\mathcal{L}[\phi, \partial\phi]$ consistent with the symmetries, keeping terms up to some power-counting criterion (“renormalizability”).
3. Quantize: canonically, or via the path integral

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}, \quad S = \int d^d x \mathcal{L}, \quad (\text{I.1.1})$$

defined in practice by expanding around the Gaussian (free) theory.

4. Renormalize: absorb the divergences of the expansion into the couplings, order by order.

The output of this procedure is a perturbation series for correlation functions, and through the LSZ reduction formula, for scattering amplitudes. The procedure’s triumphs are not in dispute. Its flagship number is the magnetic moment of the electron: the measured value $g/2 = 1.001\,159\,652\,180\,59\,(13)$ agrees with the Standard Model prediction at the level of one part in 10^{12} — the comparison is now limited by independent measurements of α , not by QFT itself.¹ The same procedure, run in reverse by Wilson, explained why critical exponents of fluids and magnets are computable at all. A definition this successful does not get discarded; it gets *relocated*, and understanding where it belongs is the business of Section I.1.5.

But look closely at what the procedure actually hands you. Step 3 writes down the symbol (I.1.1) and immediately retreats to an expansion around the quadratic action. Each order of that expansion, after step 4, is finite and unambiguous. The series as a whole is another matter, and here it pays to go to the one spacetime where every step can be done exactly: a single point.

Example I.1.1 (The ϕ^4 theory in zero dimensions)

Take spacetime to be a point. A “field” is then one real number ϕ , the “path integral” is an ordinary integral, and the Euclidean ϕ^4 partition function is

$$Z(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\phi e^{-\frac{1}{2}\phi^2 - \frac{\lambda}{4!}\phi^4}, \quad (\text{I.1.2})$$

normalized so that $Z(0) = 1$. There is nothing to renormalize and nothing to gauge-fix; the entire content of “quantizing the Lagrangian” is this one integral, which manifestly exists for all $\lambda \geq 0$.

Now run the textbook procedure: expand around the free theory. Exchanging the sum with the integral,

$$Z_{\text{pert}}(\lambda) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{4!}\right)^n \frac{1}{\sqrt{2\pi}} \int d\phi \phi^{4n} e^{-\frac{1}{2}\phi^2} = \sum_{n=0}^{\infty} a_n \lambda^n, \quad (\text{I.1.3})$$

where the Gaussian moments are $(4n-1)!! = (4n)!/(2^{2n}(2n)!)$, so that

$$a_n = \frac{(-1)^n}{(4!)^n n!} \cdot \frac{(4n)!}{2^{2n} (2n)!}. \quad (\text{I.1.4})$$

The first few terms, $Z_{\text{pert}} = 1 - \frac{\lambda}{8} + \frac{35\lambda^2}{384} - \dots$, are exactly the sums of vacuum Feynman diagrams with the usual symmetry factors. So far, so textbook. But apply Stirling’s formula

¹Fan, Myers, Sukra, Gabrielse, 2209.13084, currently the most precisely verified prediction in science. Seiberg’s essay *Thoughts About Quantum Field Theory* opens with this number for the same rhetorical purpose.

to (I.1.4):

$$|a_n| \sim \frac{1}{\pi n \sqrt{2}} n! \left(\frac{2}{3}\right)^n \quad (n \rightarrow \infty). \quad (\text{I.1.5})$$

The coefficients grow *factorially*: the ratio $|a_{n+1}/a_n| \approx \frac{2}{3}n$ diverges, and the series has zero radius of convergence. For *no* value of $\lambda \neq 0$, however small, does the perturbative expansion converge.

There is a beautifully physical way to see that this had to happen, due to Dyson. Suppose the series converged in some disc $|\lambda| < R$. Then $Z(\lambda)$ would be analytic at $\lambda = 0$ and in particular would exist for small *negative* λ . But at $\lambda < 0$ the integral (I.1.2) diverges — the potential is unbounded below, the theory has no ground state — so no such analytic continuation can exist, and $R = 0$.^a

The series is not useless — it is **asymptotic**. Truncating at order N , the error is controlled by the first omitted term, $|a_N \lambda^N| \sim N! (2\lambda/3)^N$, which *decreases* with N until $N_* \approx 3/2\lambda$ and grows explosively afterwards. Truncating at the optimal order leaves an irreducible error

$$\Delta Z \sim e^{-N_*} = e^{-3/2\lambda}, \quad (\text{I.1.6})$$

invisible at every finite order in λ . This is not a random number: the action $S = \frac{1}{2}\phi^2 + \frac{\lambda}{4!}\phi^4$ has, besides $\phi = 0$, the complex saddle points $\phi_*^2 = -6/\lambda$ with $S(\phi_*) = -3/2\lambda$, and it is these *instanton-like* saddles that govern both the large-order growth (I.1.5) and the error (I.1.6). Perturbation theory around one saddle knows about the others only through the divergence of its own coefficients.

^aDyson's original argument (Phys. Rev. **85**, 631 (1952)) was made for QED, where it is heuristic; in zero dimensions it is a rigorous one-line proof.

Three lessons, in increasing order of importance. First, the perturbation series was never the theory: it is a particular asymptotic expansion *of* the theory, tied to a particular saddle point. Second, the theory exists anyway — here, trivially, as a convergent integral. Third, the gap between the two is quantitative and structured: effects of order $e^{-3/2\lambda}$, the germ of everything “nonperturbative.” In zero dimensions these statements are almost embarrassing in their simplicity. Their force comes from dimensional promotion: in $d > 0$ the same factorial growth afflicts the loop expansion of essentially every interacting QFT, while the analogue of “the integral exists anyway” — some construction of the theory that does not lean on the expansion — becomes the entire problem. The recipe of steps 1–4 produces the series and is silent about the object the series approximates.

Remark I.1.1 (Borel resummation and resurgence). For $\lambda > 0$ the zero-dimensional series can be completely repaired: its Borel transform $B(t) = \sum_n a_n t^n / n!$ converges for $|t| < 3/2$ (the singularity sits at $t = -3/2$, the instanton action again), and the Borel integral $\int_0^\infty dt e^{-t} B(\lambda t)$ reproduces $Z(\lambda)$ exactly (Exercise I.1.3). The program of *resurgence* systematizes this: the perturbative series around each saddle encodes, through the detailed structure of its divergence, the contributions of all the others. In genuine QFTs the story is harder (renormalons join instantons among the Borel singularities) and is an active subject; see Aniceto–Başar–Schiappa [1802.10441](#) for a thorough introduction, and Part XII for where it sits today. We will not need resurgence machinery in these notes, but you should know that “the series diverges” is the beginning of a theory, not the end of one.

One should be fair about what the old framework knows beyond the series. Lattice regularization

defines Z nonperturbatively at finite cutoff, and constructive field theory has *proven* that interacting continuum limits exist in low dimensions — the $P(\phi)_2$ theories, ϕ_3^4 , Gross–Neveu₂, Yukawa_{2,3} all rigorously satisfy the Wightman axioms.² Instanton calculus patches in the leading e^{-S^*} effects. The working physicist’s position circa 2000 — “the path integral means: lattice + continuum limit, and perturbation theory is its asymptotic expansion” — is coherent and, in its domain, correct. The trouble is the domain. The next section presents four pieces of settled physics that fall outside it.

Exercise I.1.1 (easy). Verify $a_1 = -1/8$ and $a_2 = 35/384$ from (I.1.4), and check them against the vacuum Feynman diagrams of ϕ^4 theory (one and two vertices; count symmetry factors). Then verify the asymptotic statement (I.1.5) numerically: compute $|a_{n+1}/a_n|$ for $n = 5, 10, 20$ and compare with $\frac{2}{3}(n + \frac{7}{4})$.

Hint: For the diagrams: at one vertex, the figure-eight has symmetry factor 8. For the ratio: use (I.1.4) directly; no Stirling needed.

Answer: $a_{n+1}/a_n = -\frac{(4n+4)(4n+3)(4n+2)(4n+1)}{24 \cdot 4 \cdot (n+1)(2n+2)(2n+1)} = -\frac{2}{3}n - \frac{7}{6} + O(1/n)$, so the quoted ratio is accurate to a few percent already at $n = 10$.

Exercise I.1.2 (classic). (Dyson in quantum mechanics.) Consider the anharmonic oscillator $H = \frac{1}{2}p^2 + \frac{1}{2}x^2 + \lambda x^4$. Adapt Dyson’s argument to show that the Rayleigh–Schrödinger series for the ground-state energy $E_0(\lambda)$ must have zero radius of convergence. (Bender and Wu computed its coefficients to high order in 1969 and found $|c_n| \sim n!3^n$, confirming this.)

Hint: What happens to the spectrum of H for $\lambda < 0$, no matter how small? Think about tunneling out of the origin: the decay rate is $\sim e^{-c/|\lambda|}$, which is non-analytic at $\lambda = 0$.

Exercise I.1.3 (medium). (Borel summation in $d = 0$.) Define $B(t) = \sum_n a_n t^n / n!$ with a_n as in (I.1.4). Show that the series for B converges for $|t| < 3/2$, and that for $\lambda > 0$

$$\int_0^\infty dt e^{-t} B(\lambda t) = Z(\lambda).$$

Hint: Insert $1 = \frac{1}{n!} \int_0^\infty dt e^{-t} t^n$ into the (divergent) series for Z and exchange sums and integrals where legal. Convergence of B follows from (I.1.5). The Borel singularity at $t = -3/2$ sits on the *negative* axis because the series alternates — this is why the integral along the positive axis is safe. For $\lambda < 0$ the singularity obstructs the contour: the ambiguity in avoiding it is $\sim e^{-3/2|\lambda|}$, the decay rate of the metastable vacuum.

I.1.2 Four cracks

If the only deficiency of “quantize a Lagrangian” were the divergence of its series, the fix would be technical and the old definition would stand. The actual situation is more interesting: as physicists explored quantum field theory away from weak coupling — through supersymmetry, through string theory, through the bootstrap, through rigorous statistical mechanics — they found that the map between Lagrangians and theories is neither surjective nor injective, and in at least one textbook case not even well-defined. We present four exhibits. None of them is derived here; each is a vignette with pointers to the part of these notes where it is done properly, and to the literature. We flag this clearly: *in this section we cite, not derive* — which is itself a lesson in how this material is established.

²The constructive literature is reviewed in Dedushenko [2203.08053](#), §2.1, which is also the source for the axiom systems quoted later in this chapter.

Crack 1: a theory with no Lagrangian. There exists a six-dimensional interacting conformal field theory, the **(2, 0) theory**, with the following resume. It was discovered in 1995 as the decoupling limit of coincident M5-branes in M-theory, equivalently of type IIB string theory on an A–D–E singularity.³ It is, by every available diagnostic, an honest local QFT: it has a stress tensor, local operators with an operator product expansion, and correlation functions obeying the usual axioms of conformal field theory. It is maximally supersymmetric, and upon compactification on a circle it flows to five-dimensional Yang–Mills; on a torus, to four-dimensional $\mathcal{N} = 4$ Yang–Mills, whose S-duality becomes, from six dimensions, the merely *geometric* statement that a torus has a modular group. And it has no Lagrangian — not “none found yet” as a matter of bookkeeping, but structurally: the theory has no coupling constant whatsoever, no marginal or relevant deformation preserving its symmetry, hence no dial to turn to reach weak coupling, hence nothing for a path-integral expansion to expand around. Even the count of its degrees of freedom announces the problem: anomaly and holographic computations show the number of degrees of freedom of the rank- N theory grows as N^3 ,⁴ a scaling no known collection of free fields reproduces. Every property listed above was established through string dualities and consistency arguments, not through any intrinsic construction — which is precisely the scandal. Compactifying the (2, 0) theory on Riemann surfaces generates the enormous “class $\mathcal{S}$ ” of four-dimensional theories, most of which inherit its homelessness. The space of quantum field theories is strictly larger than the space of Lagrangians. (Part VI.4; Gaiotto [0904.2715](#), Tachikawa [1312.2684](#).)

Crack 2: one theory, several Lagrangians. The opposite failure is just as instructive: the map from Lagrangians to theories is many-to-one, and not only in trivial ways (field redefinitions) but in ways that exchange weak and strong coupling, bosons and fermions, gauge and global. The cleanest examples live in 2+1 dimensions. Take the $O(2)$ Wilson–Fisher fixed point — the critical point of a complex scalar with $|\phi|^4$ interaction, the universality class of the superfluid transition in ^4He . **Particle–vortex duality** asserts that the very same conformal field theory is reached by a *different* Lagrangian: a complex scalar coupled to a dynamical $U(1)$ gauge field (the abelian Higgs model), with the global symmetry that counts particles on one side realized as the symmetry that counts magnetic flux on the other,

$$|\partial_\mu \phi|^2 - |\phi|^4 \quad \longleftrightarrow \quad |D_b \hat{\phi}|^2 - |\hat{\phi}|^4 + (\text{Maxwell for } b), \quad (\text{I.1.7})$$

both tuned to criticality.⁵ Neither side is “the” description: the elementary field of one is a topological soliton of the other. In 2016 this old duality was joined by a *seed* boson–fermion duality — the gauged Wilson–Fisher boson at Chern–Simons level one is a free Dirac fermion — from which a whole web of equivalences follows by formal gauging operations, with experimental relevance from quantum Hall transitions to topological insulator surfaces. If a single conformal field theory can be presented as bosons, as gauged bosons, or as fermions, then asking “which Lagrangian is the theory?” is asking which coordinate chart is the manifold. (Part VII; [1606.01989](#).)

Crack 3: computation without a Lagrangian. The cracks above could be dismissed as exotica — supersymmetric, low-dimensional. The third cannot, because it concerns the most pedestrian

³Witten, [hep-th/9507121](#). The theory is labeled by an A–D–E Lie algebra; “the” (2, 0) theory of rank N is the A_{N-1} case.

⁴Henningson–Skenderis, [hep-th/9806087](#), from holography; Harvey–Minasian–Moore from anomaly inflow. No free-field counting reproduces N^3 .

⁵Peskin (1978) and Dasgupta–Halperin (1981) originally; stated in modern form, with the whole web it seeds, in Seiberg–Senthil–Wang–Witten [1606.01989](#), whose §1 is the recommended first reading on what duality means.

strongly-coupled QFT in nature: the critical point of the three-dimensional Ising model, the universality class of the liquid–vapor critical point and of uniaxial magnets. The **conformal bootstrap** determines its low-lying spectrum to stunning precision,

$$\Delta_\sigma = 0.5181489(10), \quad \Delta_\epsilon = 1.412625(10), \quad (\text{I.1.8})$$

— the scaling dimensions that fix the classic critical exponents η and ν — using as input *no Lagrangian, no lattice, and no expansion parameter*: only unitarity, conformal symmetry, and the associativity (crossing) of the operator algebra, implemented as a numerical exclusion problem on the space of possible CFT data.⁶ These values are more precise than any Monte Carlo simulation of any lattice realization, and incomparably more precise than the ϵ -expansion applied to the ϕ^4 Lagrangian that ostensibly “defines” the theory. Consistency conditions on the observables, taken alone, pin the theory down: the observables behave as primary data, the Lagrangian as one expendable presentation among several. How this is possible — why crossing plus unitarity is so much stronger than it looks — is the subject of Part III. (Part III.2; Rychkov’s history [2509.02779](#) tells the remarkable story, from Polyakov’s 1974 manifesto to the 2012 Ising island.)

Crack 4: the Lagrangian that defines nothing. Finally, the recipe can fail at its own game. Its standard pedagogical example — the one in which every textbook teaches renormalization — is ϕ^4 theory in $d = 4$. In 2019 Aizenman and Duminil-Copin closed a forty-year-old question by proving that the continuum limits of lattice ϕ_4^4 and of the 4d Ising model, at or near criticality, are *Gaussian*: every choice of bare couplings, however strong, yields free fields when the cutoff is removed.⁷ The interacting continuum ϕ_4^4 theory — the object steps 1–4 purport to construct — does not exist. What does exist, and what the textbook computations were always really about, is an *effective* field theory with a finite cutoff, whose couplings at low energy are insensitive to the details at the cutoff; this is also the modern understanding of the Standard Model’s scalar sector. Triviality makes the relocation unavoidable: “quantize this Lagrangian” was never a definition, but a procedure whose meaning depends on where in theory space you stand and at what resolution you look. (Part VIII.5; the $d = 4 - \epsilon$ Wilson–Fisher fixed point, which *does* exist and explains why the recipe works so well just below four dimensions, is constructed honestly in chapter I.6.)

Four cracks, one diagnosis. In each case the failure is not of quantum field theory but of the identification of a theory with its presentation. The $(2, 0)$ theory has no presentation of Lagrangian type; the $O(2)$ fixed point has several; the 3d Ising CFT is best accessed with no presentation at all; ϕ_4^4 is a presentation of nothing. The conclusion was stated by Polyakov already in 1974, in the introduction — it reads as a manifesto — of the paper that founded the conformal bootstrap: the description of a critical theory “should be ‘democratic’ with respect to the different fields,” i.e. should not privilege whatever variables happen to appear in someone’s bare Hamiltonian.⁸

⁶Numbers from Poland–Rychkov–Vichi [1805.04405](#), Table II, obtained in Simmons–Duffin [1612.08471](#); the rigorous-island determinations are by Kos–Poland–Simmons–Duffin–Vichi [1603.04436](#). Rychkov–Su [2311.15844](#) reviews the current state of the art.

⁷Aizenman–Duminil-Copin [1912.07973](#). “Marginal” triviality: $d = 4$ is the boundary case, where the proof requires a logarithmic improvement over the classic tree-diagram bounds that settled $d \geq 5$ in the 1980s. This is a theorem of mathematics, not a scenario.

⁸Polyakov, *Non-Hamiltonian approach to conformal quantum field theory*, ZhETF **66** (1974) 23; quoted and contextualized in Rychkov [2509.02779](#), §5, which reproduces the full passage. Seiberg’s one-slide summary at the 2011 Solvay meeting — no Lagrangian for some theories, several for others, magic in amplitudes, not mathematically rigorous — is the same diagnosis a generation later.

Idea I.1.1 (A theory is its observables)

A quantum field theory is the total structure of its observables — correlation functions, spectra, partition functions, extended operators — not any particular procedure for generating them. Lagrangians, lattice models, bootstrap data sets, and holographic duals are **presentations** of a theory: coordinate descriptions, each with a domain of convenience, none privileged. Some theories have many presentations; some have none we know.

This is the first organizing principle of these notes, and everything in this part flows from taking it seriously. It immediately raises the question that the rest of this chapter answers: if not a Lagrangian, then *what*? What is the structure that all presentations of a given theory present?

Preview I.1.1 (Where the cracks are repaired)

Theories without Lagrangians are constructed (by compactification and by bootstrap) in Part VI.4 and Part III. Dualities are stress-tested and partially derived in Part VII, with the tools — symmetries, anomalies, line spectra — built in Part II. The bootstrap occupies Part III. Triviality, constructive field theory, and the lattice–continuum dictionary occupy Part VIII. The question “what would a complete definition even look like?” returns in Part XII.1.

Exercise I.1.4 (thinker). Suppose a colleague proposes that two Lagrangians, like the two sides of (I.1.7), describe the same infrared theory. Neither side is solvable. List every quantity you could in principle compare *without* solving either theory. (Symmetries? Their representations on local and extended operators? Anomalies? Partition functions on nontrivial geometries? Behavior under deformations?) Keep your list; Part II is the systematic theory of exactly these duality invariants, and Part VII grades proposals against them.

Hint: The most useful invariants are the ones that cannot flow: discrete data, quantized coefficients, anything protected under the renormalization group. The modern checklist begins with the global symmetry *category* and its ‘t Hooft anomalies.

I.1.3 What a quantum field theory assigns

If a theory is not its Lagrangian, our definition-shaped question has to be approached from the other side: not “what generates the theory?” but “what does the theory *provide*?” Think of it as defining the species by its behavior. Whatever a quantum field theory is, any presentation of it must produce a definite collection of physical data, and two presentations describe the same theory exactly when all of that data agrees. The modern consensus on the inventory — assembled from the axiomatic tradition and sharpened by the functorial viewpoint — is the following.⁹

Definition I.1.1 (The inventory: data of a QFT)

A quantum field theory $\mathcal{T}$ in d spacetime dimensions assigns:

⁹This is a physicist’s distillation of structures formalized, in different fragments, by the Wightman, Osterwalder–Schrader, Haag–Kastler, and Atiyah–Segal axiom systems; see Dedushenko [2203.08053](#), §2, for all of them with full references, and Section I.1.6 below for how they compare. The geometric (“assign data to manifolds”) formulation follows Segal, recently extended by Kontsevich–Segal [2105.10161](#).

1. to a closed d -dimensional spacetime M_d (with metric, and possibly other background fields): a **partition function** $Z_{\mathcal{T}}[M_d] \in \mathbb{C}$;
2. to points $x_1, \dots, x_n \in M_d$ decorated by labels $\mathcal{O}_1, \dots, \mathcal{O}_n$ drawn from a set of **local operators**: correlation functions $\langle \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \rangle_{M_d}$;
3. to a closed $(d-1)$ -dimensional space Σ_{d-1} : a **Hilbert space** $\mathcal{H}_{\Sigma}$ of states, with a d -manifold bounded by Σ defining a state in $\mathcal{H}_{\Sigma}$ (cutting and gluing);
4. to closed submanifolds of intermediate dimension, decorated by labels from sets of **extended operators** (line operators, surface operators, $\dots$), and to boundaries and interfaces: correlation functions in their presence;

subject to consistency conditions — locality of the operators, compatibility of cutting and gluing, unitarity/reflection positivity, convergence structures for the operator product expansion — that subsequent chapters make precise.

Each entry of the inventory gets a chapter of its own (local operators in I.2, extended operators in I.3, states and gluing in I.4), so here we stay at the level of orientation, and the orientation is best gained by noticing what the list does *not* contain. There are no fields: “ ϕ ” appears nowhere, only the operators $\mathcal{O}_i$, and dualities like (I.1.7) show that what one presentation calls an elementary field another calls a composite or a soliton. There is no Lagrangian and no gauge group — the gauge “symmetry” of the abelian Higgs side of (I.1.7) is invisible in the inventory, a redundancy of that presentation rather than a property of the theory (a point developed carefully in Part V.1). And there are no particles: particles are states in $\mathcal{H}$ that happen to organize themselves into asymptotic excitations when the theory is gapped, an excellent emergent concept in some theories and simply absent in others (a conformal theory like 3d Ising has a continuous spectrum of scaling states and no particles at all). Item 4 — defects — would not have appeared on anyone’s list in 2000 and is the most consequential modern addition: theories with identical local correlators can differ in their extended operators, so the extended sector is part of the *definition*, not a garnish. That lesson gets chapter I.3 to itself.

The right way to internalize the inventory is to watch all of it materialize in a system so small that nothing can hide. Quantum mechanics is a quantum field theory in $d = 1$ — the “fields” are functions of time only — and one particular quantum mechanics is, for reasons that will keep paying off through Part V, the perfect specimen.

Example I.1.2 (The inventory in $d = 1$: a particle on a circle)

Take a free particle of unit mass on a circle, $q \simeq q + 2\pi$, with Lorentzian Lagrangian

$$L = \frac{1}{2}\dot{q}^2 + \frac{\theta}{2\pi}\dot{q}. \quad (\text{I.1.9})$$

The second term is our first **topological term**: a total derivative, with no effect on the equations of motion, whose coefficient θ will nonetheless turn out to be physical — the one-dimensional ancestor of the θ -angles of Yang–Mills theory.^a

The Hamiltonian presentation. The momentum conjugate to q is $\Pi = \dot{q} + \frac{\theta}{2\pi}$, and the

Hamiltonian is

$$H = \frac{1}{2} \left(\Pi - \frac{\theta}{2\pi} \right)^2. \quad (\text{I.1.10})$$

The inventory begins to fill in. *States* (item 3): space is a point, and the Hilbert space assigned to it is $\mathcal{H} = L^2(S^1)$, spanned by $\langle q|n\rangle = e^{inq}/\sqrt{2\pi}$ with $n \in \mathbb{Z}$ — single-valuedness of the wavefunction quantizes Π to integers. The spectrum is

$$E_n = \frac{1}{2} \left(n - \frac{\theta}{2\pi} \right)^2. \quad (\text{I.1.11})$$

Local operators (item 2): the well-defined operators are built from $e^{\pm iq}$ and Π (q itself is not single-valued, hence not an operator — already a hint that “the field” is not sacred), with correlators like $\langle 0|e^{iq(t)}e^{-iq(0)}|0\rangle$ computable by inserting energy eigenstates. *Partition function* (item 1): the closed 1-manifold is a circle; continuing to Euclidean time and traversing a circle of circumference β ,

$$Z(\beta, \theta) = \text{tr } e^{-\beta H} = \sum_{n \in \mathbb{Z}} \exp \left[-\frac{\beta}{2} \left(n - \frac{\theta}{2\pi} \right)^2 \right]. \quad (\text{I.1.12})$$

Two features of the spectrum (I.1.11) deserve a hard look. First, θ is an angle: $\theta \rightarrow \theta + 2\pi$ shifts $n \rightarrow n + 1$ and leaves the spectrum — and every entry of the inventory — invariant. (Operatorially, $H_{\theta+2\pi} = e^{iq} H_{\theta} e^{-iq}$.) Second, the invariance is realized *nontrivially*: as θ runs from 0 to 2π the levels cross, and at exactly $\theta = \pi$ the ground state is two-fold degenerate, $E_0 = E_1$. A total-derivative coupling has produced physical consequences, periodic in 2π , with a level crossing at the symmetric point that — as we will see in Parts II.6 and V.3 — no deformation respecting the symmetries can remove. This is the entire conceptual skeleton of “Yang–Mills at $\theta = \pi$,” in a system solvable by inspection.

The path-integral presentation. Now compute $Z(\beta, \theta)$ the other way, as a sum over all maps from the Euclidean time circle to the target circle. Such maps fall into disconnected topological classes labeled by the **winding number** $w \in \mathbb{Z}$; in each class the topological term contributes the *phase* $e^{i\theta w}$ — topological terms acquire no factor of i under Wick rotation and remain imaginary in Euclidean signature (Appendix A; this is the single most common sign error in the subject). The action in the class w is minimized by $q_w(\tau) = 2\pi w\tau/\beta$, with $S_E[q_w] = 2\pi^2 w^2/\beta$, and the Gaussian fluctuations around q_w are w -independent, contributing the free-particle factor $\sqrt{2\pi/\beta}$ (Exercise I.1.7). Hence

$$Z(\beta, \theta) = \sqrt{\frac{2\pi}{\beta}} \sum_{w \in \mathbb{Z}} \exp \left[-\frac{2\pi^2 w^2}{\beta} + i\theta w \right]. \quad (\text{I.1.13})$$

Expressions (I.1.12) and (I.1.13) look nothing alike — one is a sum over momenta with θ in the exponents’ shifts, the other a sum over windings with θ as a phase — yet they are equal, exactly, by the Poisson resummation formula $\sum_n f(n) = \sum_w \hat{f}(2\pi w)$ applied to the Gaussian (Exercise I.1.6). They are two presentations of one theory, and they are useful in opposite regimes: at low temperature ($\beta \rightarrow \infty$) the Hamiltonian sum is dominated by its first term or two while the winding sum converges slowly, and at high temperature ($\beta \rightarrow 0$) the roles

reverse. Keep this little duality in mind as a template: *the same observable, two expansions, each weakly coupled where the other is not*. That sentence, suitably inflated, is Part VII.

^aThis example is worked out, as a warm-up for 4d gauge theory, in Gaiotto–Kapustin–Komargodski–Seiberg 1703.00501, Appendix D.1, whose conventions we follow. It reappears in Córdova–Freed–Lam–Seiberg 1905.09315 as the simplest anomaly in parameter space.

The example is humble, but every theme of this chapter is present in it: data assigned to geometric objects (a Hilbert space to a point, a number to a circle); operators defined by the theory rather than by the classical configuration space (e^{iq} , not q); a parameter space with nontrivial topology ($\theta \in S^1$) and physics pinned to its special points; and two presentations related by a nontrivial identity. When in later chapters the going gets abstract, return here.

There is one more thing the example quietly demonstrates: at $\theta = \pi$ the classical symmetry $C : q \rightarrow -q$ combines with the shift symmetry of the free particle into a group that is realized on $\mathcal{H}$ only *projectively* — the operators implementing it commute only up to a phase. The two-fold degeneracy is the direct consequence, and the projective phase is the simplest example of a **'t Hooft anomaly**, the exactly-conserved quantity of the renormalization group that Part II builds into the subject's sharpest tool.¹⁰ We flag it and move on; it returns in II.6 with the machinery to do it justice.

Exercise I.1.5 (easy). Show directly that $U = e^{iq}$ is a well-defined unitary operator on $L^2(S^1)$, that $UH_\theta U^\dagger = H_{\theta+2\pi}$, and that U maps $|n\rangle \rightarrow |n+1\rangle$. Conclude that all inventory data is 2π -periodic in θ , and identify which states cross at $\theta = \pi$.

Hint: Work in the $|n\rangle$ basis throughout; e^{iq} is a shift operator on momentum eigenstates.

Exercise I.1.6 (classic). Prove the Poisson resummation formula $\sum_{n \in \mathbb{Z}} f(n) = \sum_{w \in \mathbb{Z}} \hat{f}(2\pi w)$, $\hat{f}(k) = \int dx f(x) e^{-ikx}$, and use it to derive (I.1.13) from (I.1.12), including the prefactor and the sign of the phase.

Hint: The Dirac comb $\sum_n \delta(x-n)$ is periodic, so it equals its own Fourier series; integrate f against both expressions. For the application: complete the square in (I.1.12) first.

Answer: With $f(x) = e^{-\frac{\beta}{2}(x-\frac{\theta}{2\pi})^2}$ one finds $\hat{f}(2\pi w) = \sqrt{2\pi/\beta} e^{-2\pi^2 w^2/\beta} e^{-i\theta w}$; relabeling $w \rightarrow -w$ gives (I.1.13).

Exercise I.1.7 (medium). Justify the prefactor in (I.1.13): show that the Gaussian integral over periodic fluctuations $\delta q(\tau)$ around any winding saddle is independent of w , and that the zero mode (the constant shift of q) produces a factor of the target-circle circumference, yielding in total the free-particle result $2\pi/\sqrt{2\pi\beta} = \sqrt{2\pi/\beta}$.

Hint: $S_E[q_w + \delta q] = S_E[q_w] + \frac{1}{2} \int \dot{\delta q}^2$ exactly — the action is quadratic and the cross term integrates to zero by periodicity. For the normalization, compare with the diagonal heat kernel of a free particle on a line, $\langle q|e^{-\beta H}|q\rangle = 1/\sqrt{2\pi\beta}$.

The particle on a circle entered through a Lagrangian, however — hardly a clean break from the old paradigm. So let us fill the same inventory from the opposite end of physics: a statistical lattice with no Lagrangian, no continuum, and at first sight no quantum mechanics at all.

¹⁰GKKS 1703.00501, eqs. (D.5)–(D.10): the symmetry group at $\theta = \pi$ is a central extension of $O(2)$ — $\text{Pin}(2)$ — and a $\text{Pin}(2)$ that cannot be reduced to $O(2)$ forces degenerate pairs.

Example I.1.3 (The Ising chain, read as a field theory)

Put classical spins $s_i = \pm 1$ on N sites of a periodic chain, with energy favoring alignment, and define

$$Z_N = \sum_{\{s_i=\pm 1\}} \prod_{i=1}^N e^{Js_i s_{i+1}}, \quad s_{N+1} \equiv s_1, \quad (\text{I.1.14})$$

where $J > 0$ absorbs the temperature. This is a Euclidean QFT presentation in which spacetime itself is discrete: the chain is a one-dimensional “spacetime,” (I.1.14) its partition function on a (discrete) circle.

The structure of the inventory emerges from a single move. Define the 2×2 **transfer matrix** with entries $T_{ss'} = e^{Js s'}$,

$$T = \begin{pmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{pmatrix} = e^J \mathbb{1} + e^{-J} \sigma_x, \quad \text{eigenvalues } \lambda_{\pm} = 2 \cosh J, 2 \sinh J. \quad (\text{I.1.15})$$

Multiplying along the chain and summing over spins,

$$Z_N = \text{tr } T^N = \lambda_+^N + \lambda_-^N. \quad (\text{I.1.16})$$

Compare item by item with the previous example. The *Hilbert space* assigned to a slice of this spacetime — a single site — is $\mathbb{C}^2$, the span of $|\uparrow\rangle, |\downarrow\rangle$. The *Euclidean time evolution* over one lattice step is T , the discrete stand-in for e^{-aH} . The *partition function on a circle* is a trace, exactly as $\text{tr } e^{-\beta H}$ was. And the *correlators*: inserting σ_z at sites 0 and r ,

$$\langle s_0 s_r \rangle = \frac{\text{tr}(\sigma_z T^r \sigma_z T^{N-r})}{\text{tr } T^N} \xrightarrow{N \rightarrow \infty} \left(\frac{\lambda_-}{\lambda_+} \right)^r = (\tanh J)^r = e^{-r/\xi}, \quad (\text{I.1.17})$$

with **correlation length** $\xi = 1/\ln \coth J$ (Exercise I.1.9). Two general morals hide in this formula. The correlation length is the inverse of the *gap*: $1/\xi = \ln(\lambda_+/\lambda_-)$ is the spacing of the transfer matrix’s spectrum, the discrete avatar of $m = E_1 - E_0$. And **cluster decomposition** — correlations factorize at long distance, $\langle s_0 s_r \rangle \rightarrow \langle s \rangle^2$ — is on explicit display: it is nothing but the statement that the largest eigenvalue dominates.

Now take the *continuum limit*, the move that will occupy all of Part VIII; here it can be done exactly. Give the lattice a physical spacing a and send $a \rightarrow 0$, $N \rightarrow \infty$. For the limit to retain any physics, the correlation length *in lattice units* must diverge, so $J \rightarrow \infty$ too; the combination to hold fixed is the physical gap. Writing $T = e^J (\mathbb{1} + e^{-2J} \sigma_x + \dots)$ and matching to $T \simeq e^{-aH + \text{const}}$,

$$H = -\Gamma \sigma_x, \quad \Gamma = \frac{e^{-2J}}{a} \text{ held fixed as } a \rightarrow 0. \quad (\text{I.1.18})$$

The continuum limit of the classical Ising chain is the quantum mechanics of a single spin in a transverse field — Hilbert space $\mathbb{C}^2$, gap 2Γ , physical correlation length $\xi_{\text{phys}} = a\xi \rightarrow 1/2\Gamma$. Every microscopic detail (the lattice, the specific nearest-neighbor form, corrections to the exponentiation) has washed out into one parameter. This is **universality** in its smallest habitat, and it is the engine that makes “QFT from a lattice” meaningful at all: the continuum theory remembers only the data that survives the limit.

Two systems could hardly look more different than a quantum particle on a circle and a classical

chain of spins, yet they filled the same inventory in the same pattern: Hilbert space on a slice, (Euclidean) evolution operator, trace on a circle, correlators with definite decay. That pattern — not any Lagrangian — is what Definition I.1.1 axiomatizes. The chain also displays, in miniature, the claim from the chapter opening that the standard paradigm gets *relocated* rather than discarded: “quantize a Lagrangian” lives inside the inventory picture as one family of presentations (start from classical configurations, build T or e^{iS} , read off the data), valid whenever it converges on something — and the chain shows there are presentations, perfectly good ones, that never mention a classical action in the continuum.

One disclaimer, before the lattice picture settles in too comfortably as “the” nonperturbative definition. In $d = 1$ the continuum limit (I.1.18) is a theorem of elementary analysis. In higher dimensions, “the continuum limit exists and is universal” is a theorem only in special cases, a well-supported conjecture in most, and — this is the surprise of the 2020s — *false* in instructive ones, where lattice models produce continuum descriptions that violate cherished axioms (fractons, Part VIII.4) or where the limit exists but is free despite all appearances (triviality, crack 4). The lattice is a presentation too, with its own dictionary problems; Part VIII is the audit.

Exercise I.1.8 (easy). Add a magnetic field: $Z_N = \sum_{\{s\}} e^{\sum_i (J s_i s_{i+1} + h s_i)}$. Find the symmetric transfer matrix, its eigenvalues, and the magnetization $m = \langle s_i \rangle$ in the thermodynamic limit.

Hint: Use $T_{ss'} = e^{J s s' + \frac{h}{2}(s+s')}$ to keep T symmetric. Then $m = \partial \ln \lambda_+ / \partial h$ at fixed J .

Answer: $\lambda_{\pm} = e^J \cosh h \pm \sqrt{e^{2J} \sinh^2 h + e^{-2J}}$, giving $m = \sinh h / \sqrt{\sinh^2 h + e^{-4J}}$: smooth for every $J < \infty$, with no spontaneous magnetization at $h = 0$.

Exercise I.1.9 (classic). Derive (I.1.17): insert σ_z operators into the trace, evaluate in the eigenbasis of T , and keep track of the finite- N corrections. Verify cluster decomposition and extract ξ .

Hint: σ_z is off-diagonal in the eigenbasis $|\pm\rangle = (|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$: $\sigma_z |\pm\rangle = |\mp\rangle$. The exact answer is $(\lambda_+^{N-r} \lambda_-^r + \lambda_-^{N-r} \lambda_+^r) / (\lambda_+^N + \lambda_-^N)$.

Exercise I.1.10 (medium). Make the continuum limit (I.1.18) precise. Show that the corrections to $T \approx e^J e^{-aH}$ are of relative order $e^{-4J} = (a\Gamma)^2$ and hence vanish in the limit; show that the limit of N -site quantities with $Na = \beta_{\text{phys}}$ fixed reproduces the spin- $\frac{1}{2}$ thermal traces; and verify $\xi_{\text{phys}} \rightarrow 1/2\Gamma$ from (I.1.17).

Exercise I.1.11 (thinker). The chain never orders: Exercise I.1.8 shows the free energy is analytic in (J, h) for all finite J . Give two proofs that no phase transition can occur in the 1d Ising model with short-range interactions: (a) algebraic — the transfer matrix has strictly positive entries; invoke the Perron–Frobenius theorem to show $\lambda_+ > \lambda_-$ always, so the gap never closes; (b) physical — estimate the free energy of a single domain wall in a chain of length L and show entropy always wins (Peierls’ argument, run in reverse). What do these arguments use about $d = 1$, and which step fails in $d = 2$?

Hint: For (b): a wall costs fixed energy $2J$ but can sit at any of $\sim L$ positions, so $\Delta F = 2J - T \ln L \rightarrow -\infty$. In $d = 2$ a wall is a line whose energy *and* entropy both scale with its length — the competition becomes a genuine fight, and Peierls showed order wins at low temperature.

I.1.4 Locality, the one non-negotiable

The inventory of Definition I.1.1, as stated, is satisfied by systems nobody would call a field theory — nothing yet distinguishes the Ising chain from a quantum computer’s register or from a single harmonic oscillator, which also assigns Hilbert spaces and thermal traces. What makes a theory

a theory *of fields* is not the size of its Hilbert space but the way its observables are organized by *geometry*: degrees of freedom are distributed over space, and they interact only with their neighbors. Every word of “quantum field theory” except “locality” is negotiable — we have already negotiated away the fields — and the modern attitude is to treat locality as the actual subject. QFT is what you get when many degrees of freedom are organized locally; everything else is presentation.

This one idea wears different uniforms in different formulations, and learning to recognize it in all of them is part of the training:

- *In the operator content*: operators are labeled by the points (and submanifolds) where they live, and the product of two nearby local operators is again a local operator — the **operator product expansion**, the subject of chapter I.2.
- *In Lorentzian signature*: measurements at spacelike separation are independent, $[\mathcal{O}_1(x), \mathcal{O}_2(y)] = 0$ for $(x - y)^2 > 0$ — **microcausality** (Exercise I.1.12).
- *In Euclidean correlators*: **cluster decomposition**, already met in (I.1.17) — distant regions decorrelate.
- *In the dynamics*: the Hamiltonian is an integral of a local density. Better, because it is presentation-independent: the theory possesses a local **stress tensor** $T_{\mu\nu}(x)$ — the operator that measures energy and momentum density, and the response of the theory to deforming the geometry it lives on. The stress tensor is the one operator every local QFT shares, and its existence is the sharpest single criterion for “is this a local field theory?” (Chapter I.2 develops it; Part IX makes it the protagonist.)
- *In the path integral and on the lattice*: the action/energy is a sum of terms each involving only nearby points — nearest-neighbor couplings in (I.1.14) — which is what makes *cutting* spacetime along a slice meaningful and gives the gluing rules of item 3 of the inventory their content (chapter I.4).

These are not independent axioms but facets of one property, and their precise interrelation (which implies which, under what hypotheses) is genuinely subtle — it is the content of the reconstruction theorems discussed in chapter I.2. For orientation, the slogan suffices:

Idea I.1.2 (Locality is the subject)

A quantum field theory is a quantum system whose observables are organized by the geometry of spacetime, with independence at large separation and a local stress tensor as the operator witness. Locality is the one ingredient of the textbook definition that survives unweakened into every modern formulation — and the boundary of QFT as a subject traces exactly where locality starts to fail.

The final clause earns its keep: the failure modes are sharp and instructive. A quantum system with all-to-all couplings — mean-field models, or the SYK model beloved of holographers — has no notion of spatial distance and is quantum mechanics, not field theory (often a useful caricature of one, which is its point). More exotically, the *little string theories* arising on NS5-branes have local energy *density* in a loose sense but provably no stress tensor and no local operators creating their excitations; they sit just outside the boundary, and Part XII.3 visits them as instructive

aliens. And systems with power-law interactions, like the long-range Ising model $J_{ij} \sim |i - j|^{-1-s}$, interpolate: for s large enough they flow to ordinary local field theories, for small s to genuinely nonlocal continuum theories, with a phase boundary in between — locality of the IR theory is itself a dynamical question.

Note I.1.1. Locality should not be confused with *Lorentz invariance*, its frequent companion. Nonrelativistic, anisotropic, and Lifshitz-type field theories are perfectly local, organized by geometry, and within our subject (condensed matter supplies them in bulk); they simply weight time and space asymmetrically. The axiom systems of Section I.1.6, built around the Poincaré group, mostly exclude them — one of several signs that the existing axioms draw the boundary in the wrong place. Fractons (Part VIII.4) push on exactly this seam.

Why promote locality, rather than, say, unitarity or symmetry, to the definition’s one fixed point? Because locality is what gives quantum field theory its unreasonable universality — the empirical fact, which the next section turns into a picture, that the same continuum theories appear as descriptions of magnets, fluids, metals, the QCD vacuum, and the early universe. The mechanism is the renormalization group: whatever the microscopic degrees of freedom, so long as they are many and locally coupled, viewing them at longer and longer wavelengths erases microscopic detail in a controlled way, and the possible long-distance destinations are heavily constrained precisely *because* locality survives the coarse-graining. Wilson’s insight, restated in the vocabulary of this chapter: QFT is not fundamentally a theory of elementary particles; it is the universal effective description of locally-organized many-body systems, of which elementary particle physics is one (spectacular) instance.¹¹

Exercise I.1.12 (classic). (Microcausality for the free scalar.) In the free massive scalar theory in $d = 4$ with mostly-plus signature, the commutator of fields is a c-number, $[\phi(x), \phi(y)] = \Delta(x - y) \mathbb{K}$. Compute Δ from the mode expansion and prove it vanishes whenever $x - y$ is spacelike.

Hint: $\Delta(x - y) = \int \frac{d^3 p}{(2\pi)^3 2E_p} (e^{ip \cdot (x-y)} - e^{-ip \cdot (x-y)})$ with $p \cdot x = -Et + \vec{p} \cdot \vec{x}$ in mostly-plus. For spacelike separation there is a Lorentz frame with $x^0 = y^0$; show the integrand is then odd under $\vec{p} \rightarrow -\vec{p}$. Why does this argument fail (as it must) for timelike separation?

I.1.5 Theory space: the picture to keep

We can now assemble the chapter’s deliverable: the picture within which all twelve parts of these notes take place. It needs three ingredients, all already on the table — the inventory (a theory is a consistent assignment of observables), locality (the assignment is organized by geometry), and universality (microscopically different systems share long-distance descriptions).

Imagine, then, the set of all quantum field theories in d dimensions: every consistent inventory, whether or not anyone has presented it. Call it **theory space**. A point of theory space is a theory — the 3d Ising CFT is a point, the free photon is a point, QCD with physical quark masses is a point. A Lagrangian is not a point: it is a *name* for a point (and dualities are the statement that points have several names). This space is the arena. Its geography is given by the **renormalization group**: inspecting a theory at successively longer distances — integrating out, coarse-graining, “zooming

¹¹Seiberg’s *Thoughts About QFT* presses this point: “quantum field theory is the modern calculus — the natural language for describing diverse phenomena.” The high-energy and condensed-matter views of what a QFT is are compared, as two charts on one space of theories, in Section I.1.5.

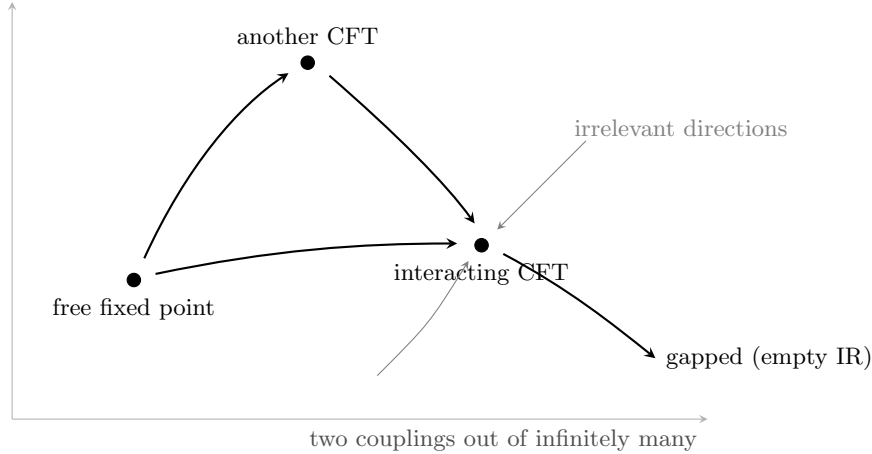


Figure 1: Theory space, schematically. Points are theories; the renormalization group moves every non-fixed theory toward longer distances along the arrows; fixed points (dots) are scale-invariant theories — CFTs — and serve as the landmarks from which the rest of the space is charted. Flows out of a fixed point follow its relevant directions (finitely many); the gray arrows are irrelevant directions, along which flows converge *into* a fixed point, producing universality.

out” — produces a one-parameter motion through theory space,

$$\frac{d}{d \ln(\text{scale})} (\text{theory}) = \beta(\text{theory}), \quad (\text{I.1.19})$$

a flow, of which chapter I.6 gives the honest construction. Here we need only its fixed-point structure, and the picture in Figure 1.

A **fixed point** of the flow is a theory invariant under zooming out: a scale-invariant theory. In all known unitary local examples scale invariance comes enhanced to conformal invariance,¹² so the landmarks of theory space are **conformal field theories** — theories with no intrinsic length scale, looking the same at every magnification. The 3d Ising CFT of crack 3 is precisely such a landmark, which is why it could be studied with no reference to any flow that reaches it. It is worth pausing on how recent the consensus behind this picture is: when Polyakov suggested at a 1970 conference that critical points possess scale (indeed conformal) invariance, both C. N. Yang and T. T. Wu rose to object that scale invariance in statistical mechanics was neither mathematically nor physically justified.¹³

Near a fixed point, the flow linearizes, and the eigendirections sort themselves by their behavior under zooming out: **relevant** perturbations grow (they carry you away from the fixed point), **irrelevant** ones shrink (flows converge back), **marginal** ones sit at the threshold. This tangent-space structure is computable the moment you know the scaling dimensions of the fixed point’s operators. At the free (Gaussian) fixed point of a single scalar, demanding that the kinetic action $\int d^d x (\partial\phi)^2$ be dimensionless fixes

$$[\phi] = \frac{d-2}{2}, \quad \text{so that} \quad \left[\int d^d x g_n \phi^n \right] = 0 \implies [g_n] = d - n \frac{d-2}{2}. \quad (\text{I.1.20})$$

¹²Proven in $d = 2$ (Zamolodchikov–Polchinski), established under mild assumptions in $d = 4$, and true in every known unitary example; see Nakayama’s review 1302.0884. We will use “fixed point” and “CFT” interchangeably, flagging the (conjectural) gap when it matters.

¹³The exchange is transcribed in Rychkov 2509.02779, §2 — worth reading in full as a calibration of how non-obvious the now-obvious can be.

The perturbation $g_n\phi^n$ is relevant when $[g_n] > 0$, i.e. $n < 2d/(d-2)$: in $d = 4$ the relevant interactions stop at ϕ^3 with ϕ^4 marginal; in $d = 3$, at ϕ^5 with ϕ^6 marginal; in $d = 6$, already ϕ^3 is the marginal frontier; and in $d = 2$, where $[\phi] = 0$, *every* potential is relevant — the first hint that two dimensions is a world of its own (Part III.4). Operators with more derivatives, like $(\partial\phi)^4$, are irrelevant in all $d \geq 2$. The pattern generalizes beyond free fixed points — at an interacting CFT the same counting runs on the true scaling dimensions Δ , with “relevant” meaning $\Delta < d$ — and the crucial finiteness persists: *a fixed point has finitely many relevant directions*.

That finiteness demystifies, in two sentences, the step of the standard recipe that was always presented as a mystery. “Renormalizable interactions” are exactly the relevant and marginal directions at the Gaussian fixed point: the textbook instruction *write only renormalizable terms* is the geometric statement *parametrize the flows leaving this fixed point*, and the miracle that infinitely many possible counterterms reduce to a few measurable couplings is no miracle — the irrelevant directions die under the flow, so the IR physics could never have depended on them. Conversely, every QFT used in physics is an **effective field theory**: a local chart around a fixed point, accurate up to corrections suppressed by powers of the distance scale, with the irrelevant operators recording — in a decaying, organized hierarchy — the memory of whatever lies upstream. The expanded version of these claims, with the Wilson–Fisher fixed point in $d = 4 - \epsilon$ constructed as the first nontrivial landmark, is chapter I.6’s job.

The same picture finally lets us say sharply what was right about the two traditional answers to “what is a QFT?”, which can now be seen as two *charts* on theory space.¹⁴ The high-energy chart starts at a fixed point and deforms by its finitely many relevant and marginal couplings (plus, implicitly, infinitesimal irrelevant ones): a small, rigid parametrization, anchored on a landmark — this is the Lagrangian paradigm, properly understood. The condensed-matter chart starts from a lattice model at the cutoff and allows *arbitrary* local couplings and arbitrary added degrees of freedom: an enormous, floppy parametrization, in which the natural questions are about the possible long-distance *phases* and the walls between them. The two charts overlap where both converge, and translating between them — which lattice models flow to which continuum landmarks, and what data survives the trip — is a running concern of these notes, climaxing in Part VIII.

Remark I.1.2 (How much structure does theory space have?). It is tempting to ask global questions: is theory space connected? What are its components? A sharp version uses the notion of a **deformation class**: two theories are in the same class if one can interpolate between them by adding degrees of freedom at high energy and moving along couplings, preserving the symmetries one cares about. Seiberg has conjectured that the deformation class is determined by precisely two pieces of data — the global symmetry and its ’t Hooft anomalies — both of which are exactly preserved along any flow. If true, this gives theory space a remarkably simple coarse topology, with the anomalies of Part II as its complete invariants. The conjecture motivates much current work; we will be in a position to appreciate its content (and its supersymmetric stress tests) after Part II. For now it is a preview of how seriously the picture of this section can be taken: theory space is not a metaphor but an object with topology, invariants, and conjectures of its own.

The rest of these notes is organized as questions about Figure 1. What structures are transported *exactly* along the flow lines, surviving from any presentation down to the deep infrared? (Symmetries

¹⁴This comparison is drawn from Seiberg’s *Thoughts About QFT*, whose slides phrase it as “QFT in high-energy theory” vs. “QFT in condensed-matter physics.”

and their anomalies: Part II.) What are the fixed points, studied as intrinsic objects? (Conformal field theory: Part III.) Which flows between landmarks are possible, and is the flow irreversible? (Monotonicity theorems and infrared constraints: Part IV.) How do gauge theories, the solvable supersymmetric corners, and the duality webs populate the map? (Parts V–VII.) When does a lattice presentation actually converge to a point of this space? (Part VIII.) The picture is simple; its load-bearing capacity is the discovery of the last twenty years.

Exercise I.1.13 (easy). Tabulate $[\phi]$ and the relevant/marginal/irrelevant verdicts for ϕ^n ($n \leq 6$) and $(\partial\phi)^4$ in $d = 2, 3, 4, 6$. For a Dirac fermion (kinetic term $\int d^d x i\bar{\psi}\not{\partial}\psi$), find $[\psi]$ and classify the four-fermion interaction $(\bar{\psi}\psi)^2$ in $d = 2, 3, 4$. Which famous theories live at the marginal frontier in each d ?

Answer: $[\psi] = (d-1)/2$, so $[(\bar{\psi}\psi)^2] = 2(d-1)$ and the coupling has dimension $2-d$: marginal in $d = 2$ (Thirring, Gross–Neveu), irrelevant in $d > 2$ — which is why the 4d Fermi theory of weak interactions demanded a UV completion.

Exercise I.1.14 (medium). Check (I.1.20) against an honest correlator: compute the Euclidean two-point function of a free massless scalar, $\langle\phi(x)\phi(0)\rangle = \frac{C_d}{|x|^{d-2}}$, from $\int d^d p \frac{e^{ip\cdot x}}{p^2}$, and confirm that the power of $|x|$ matches $2[\phi]$ with $[\phi] = (d-2)/2$. What replaces the power law in $d = 2$, and how does this connect to the “every potential is relevant” verdict above?

Hint: In $d = 2$ the integral is IR-divergent and the two-point function grows logarithmically: $\langle\phi\phi\rangle \sim -\frac{1}{2\pi} \ln|x|$. A dimensionless field is on the edge — whence both the richness (vertex operators $e^{i\alpha\phi}$ of continuously variable dimension) and the delicacy (no SSB of continuous symmetries) of $d = 2$.

Exercise I.1.15 (thinker). Nothing in this section’s picture forbids closed RG trajectories: flows that leave a fixed point and return, or limit cycles with no fixed point at all. Reflect on what a limit cycle would mean physically (physics periodic in the *logarithm* of the scale — discrete scale invariance) and why one might expect coarse-graining, which discards information, to be gradient-like instead. This intuition is correct for unitary theories in even dimensions ≤ 4 , where it is a theorem (c - and a -theorems, Part IV.1) — but the proofs need real input beyond “information is lost,” and non-unitary counterexamples exist.

I.1.6 What would count as an answer?

The question in this chapter’s title has, today, no complete answer — but the partial answers are substantial, and a reader should know the shape of the existing attempts before chapters I.2–I.4 borrow structure from all of them. Each of the four main axiomatic traditions formalizes a different row of the inventory of Definition I.1.1.¹⁵

The **Wightman axioms** (1950s) and their Euclidean counterpart, the **Osterwalder–Schrader axioms** (1970s), axiomatize item 2: fields as operator-valued distributions, correlators with specified analyticity, positivity, and causality properties, and reconstruction theorems that rebuild the Hilbert space from the correlators. Within their domain they are a complete success — the spin-statistics and CPT theorems are their fruits, and the constructive program built genuine interacting examples in $d = 2, 3$ satisfying them. Their domain, however, is local operators in flat space: gauge theories fit awkwardly (the gauge-variant fields are not Wightman fields, and the physical content includes the extended operators of item 4, about which the axioms are silent), and the $(2, 0)$ theory — with no fields at all in any classical sense — fits not at all.

¹⁵This entire section compresses Dedushenko’s survey [2203.08053](#), which should be consulted for precise statements and for the hundred-some references the one-line summaries below conceal. The comparison returns, with the benefit of the whole book’s examples, in Part XII.1.

Algebraic QFT (Haag–Kastler, 1960s onward) drops fields in favor of assigning to each spacetime *region* the algebra of observables measurable inside it, with compatibility under inclusion and commutativity at spacelike separation. It is the formulation in which the deep structural results about quantum information in field theory live — the type classification of local algebras, Tomita–Takesaki modular theory, the Reeh–Schlieder theorem — and Part IX of these notes is, in effect, its payoff. Its classic limitation is mirror to Wightman’s: theories are easy to axiomatize but hard to construct, and topological sectors (instantons, again the extended operators) strain the axioms’ locality assumptions in ways that current research is still sorting out.

Functorial QFT (Atiyah–Segal, late 1980s; Kontsevich–Segal for the non-topological case) axiomatizes items 1 and 3 directly: a QFT is an assignment of Hilbert spaces to space-slices and of evolution maps to spacetime cobordisms between them, compatible with gluing — the path integral’s cutting-and-sewing behavior elevated to the definition. For *topological* theories this definition is not merely rigorous but spectacularly classifiable, and it is the natural home of every “assign data to manifolds” instinct in this chapter (chapter I.4 is built on it). For theories with metric dependence it faces real open problems: which complex metrics are allowable (the Kontsevich–Segal criterion [2105.10161](#), chapter I.4’s modern answer to “what is Wick rotation?”), and how to accommodate the scheme ambiguities of genuine QFTs, where the partition function on a curved background is not a number but a number-up-to-local-counterterms.

And the **lattice** tradition — physics’ working definition — defines a theory as the continuum limit of a regularized system, with constructive field theory as its rigorous wing. Its strengths and its audit findings (triviality; exotic limits) have already appeared in this chapter, and Part VIII is devoted to it.

Four traditions, four fragments — and the honest summary is that none captures class $\mathcal{S}$, large- N limits, *and* fracton models simultaneously, while the developments of Parts II–XI (symmetry categories as part of a theory’s identity; defects on equal footing with local operators) are only beginning to be reflected in any of them. Dedushenko’s verdict stands: *formulating a complete definition of QFT is an outstanding open problem*, of equal interest to physics and mathematics. These notes take the working position already staked out in this chapter — the inventory, locality, and consistency, used opportunistically through whatever presentation converges — and treat the missing definition not as an embarrassment but as the subject’s standing research frontier, revisited with better equipment in Part XII.1.

Note I.1.2 (What these notes assume). Operationally, the chapters ahead assume of every theory: unitarity (reflection positivity in Euclidean signature), a local stress tensor, cluster decomposition, and — in flat space — Poincaré invariance, flagging explicitly each time one of these is relaxed (non-unitary fixed points, little strings, fractons, theories without stress tensors all make scheduled appearances). When an argument depends on an unproven structural assumption — existence of a continuum limit, convergence of an OPE beyond CFT, enhancement of scale to conformal invariance — the dependence is stated where it occurs.

Chapter exercises

Exercise I.1.16 (classic). (Two Lagrangians, one theory, in $d = 0$.) In the integral (I.1.2), change variables $\phi = \psi + c\psi^3$ for small constant c . Show that the resulting “theory of ψ ” has a different-looking action — new interaction vertices $\psi^4, \psi^6, \dots$ with c -dependent couplings, plus a vertex from the Jacobian — yet *identical* partition function, and (after matching sources) identical correlators

of the original operator. Conclude that coupling constants of presentations are not, in general, observables. Which combinations of the new couplings are invariant?

Hint: The Jacobian $d\phi = (1 + 3c\psi^2) d\psi$ exponentiates into the action as $-\ln(1 + 3c\psi^2)$. This freedom of field redefinition is the $d = 0$ shadow of the equivalence theorem; “the theory” is the equivalence class.

Exercise I.1.17 (medium). (Large orders from the complex saddle.) Represent the coefficients of Example I.1.1 as $a_n = \oint \frac{d\lambda}{2\pi i} \frac{Z(\lambda)}{\lambda^{n+1}}$ and evaluate the large- n behavior by saddle point: the contour is pushed onto the negative- λ cut, where the discontinuity of Z is computed by the instanton saddle $\phi_*^2 = -6/\lambda$ with action $-3/2\lambda$. Recover the growth $|a_n| \sim n! (2/3)^n$ of (I.1.5), including the power of n in the prefactor if you are feeling thorough.

Hint: Disc $Z(-|\lambda|) \propto e^{-3/2|\lambda|}/\sqrt{|\lambda|}$ from the Gaussian integral over the unstable direction at the saddle. Then $a_n \approx \frac{1}{\pi} \int_0^\infty \frac{d|\lambda|}{|\lambda|^{n+1}} \text{Im} Z(-|\lambda|)$ is dominated by $|\lambda| \sim 3/2n$. This dispersive route from instantons to large orders is the $d = 0$ version of the Lipatov/Bender–Wu analysis.

Exercise I.1.18 (medium). (A correlator on the circle.) For the particle on a circle at general θ , compute the Euclidean two-point function $\langle e^{iq(\tau)} e^{-iq(0)} \rangle_\beta$ by inserting energy eigenstates, and examine its $\beta \rightarrow \infty$ limit. Show that the answer decays exponentially in τ with rate set by the gap, in parallel with (I.1.17) — except at $\theta = \pi$, where the degeneracy makes part of the correlator *persist* at infinite separation. Interpret the persistent piece as spontaneous breaking of the C symmetry between the two ground states, in the only way quantum mechanics permits: exact degeneracy.

Hint: $e^{iq}|n\rangle = |n+1\rangle$, so the correlator is $\frac{1}{Z} \sum_n e^{-(\beta-\tau)E_n - \tau E_{n+1}}$. At $\theta = \pi$, the $n = 0$ term contributes $e^{-\beta E_0}$ with no τ -dependence in the exponent’s leading piece, since $E_0 = E_1$.

Exercise I.1.19 (thinker). Each of the four cracks of Section I.1.2 embarrasses at least one axiom system of Section I.1.6. Match them: which axioms could in principle accommodate the $(2,0)$ theory but cannot construct it? Which formalize a duality like (I.1.7) as an exact statement, and which cannot even state it? Which tradition does the triviality theorem vindicate, and which does it humble? Use Dedushenko [2203.08053](#) §3 as a referee for your answers.

Exercise I.1.20 (thinker). Idea I.1.1 says a theory is its observables. Here is a stress test you can hold in mind through chapter I.3: could two *distinct* theories have identical correlation functions of all *local* operators on $\mathbb{R}^d$, differing only in their extended operators or in their partition functions on topologically nontrivial spacetimes? (No proof expected — but formulate precisely what data would have to differ, and what experiment, in a finite-size sample, would tell the two theories apart.) The answer is yes, the canonical examples being gauge theories with the same algebra and different global structure, e.g. $SU(2)$ vs. $SO(3)$ Yang–Mills; this is the content of “reading between the lines” [1305.0318](#), and the reason item 4 sits in Definition I.1.1.

Reading

Dedushenko, *The Quest to Define QFT*, [2203.08053](#). The survey behind Section I.1.6: every axiom system, with its successes and its stress fractures, in 30 readable pages plus a heroic bibliography. Read §§1, 3 now and §2 as reference.

Rychkov, *Conformal bootstrap: from Polyakov to our times*, [2509.02779](#). The history that makes the modern viewpoint feel inevitable rather than fashionable: the 1970 Kyiv exchange, Polyakov’s 1974 manifesto, and the road to the Ising island. Sections 2–5 complement this chapter; the rest seeds Part III.

Seiberg, *Thoughts About Quantum Field Theory* (IAS). The reorganization manifesto in slide form: theories without Lagrangians, deformation classes, symmetries-and-anomalies as the invariants of theory space. Twenty minutes that frame the whole book.

Gaiotto–Kapustin–Komargodski–Seiberg, [1703.00501](#), Appendix D. The particle on a circle, done as a warm-up for Yang–Mills at $\theta = \pi$ — the source for Example I.1.2 and the bridge to Parts II.6 and V.3.

Seiberg–Senthil–Wang–Witten, [1606.01989](#), §1. What duality means, by example; the duality web behind crack 2.

Aizenman–Duminil-Copin, [1912.07973](#), §1. The triviality theorem. The introduction is readable by physicists and states exactly what is and is not proven.

Poland–Rychkov–Vichi, [1805.04405](#), §1. The data-centric view of CFT from the practitioners; the source of the numbers in (I.1.8).

Books (for the shelf, not the week): Glimm–Jaffe, *Quantum Physics: A Functional Integral Point of View* — what “constructing a QFT” means when done in earnest; Weinberg, *The Quantum Theory of Fields I*, ch. 1 — the historical case that QFT was forced on us, the perfect foil for this chapter’s question.

A Conventions

Units and signature. We set $\hbar = c = 1$ throughout. The Lorentzian metric signature is **mostly-plus**, $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ (and its obvious truncations in other dimensions). Consequently a free scalar has

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2, \quad p^2 = p_\mu p^\mu = -m^2 \text{ on shell.} \quad (\text{A.1})$$

Srednicki and Weinberg use this signature; Peskin–Schroeder does not. Much of the symmetry and CFT literature works directly in Euclidean signature, where the issue is moot.

Wick rotation. Euclidean time is $t = -i\tau$ with τ real, so that $e^{iS_L} = e^{-S_E}$ and a standard kinetic term rotates as

$$i \int dt \frac{1}{2} \dot{q}^2 \longrightarrow - \int d\tau \frac{1}{2} \left(\frac{dq}{d\tau} \right)^2, \quad (\text{A.2})$$

i.e. S_E is positive for a standard kinetic term. Thermal physics at temperature $T = 1/\beta$ lives on a Euclidean time circle of circumference β : $Z(\beta) = \text{tr } e^{-\beta H}$.

Topological terms stay imaginary. A term in the action with no dependence on the metric acquires no factor of i under Wick rotation: it enters the Euclidean weight as a pure phase. For the θ -term of quantum mechanics on a circle (chapter I.1),

$$i \frac{\theta}{2\pi} \int dt \dot{q} \longrightarrow i \frac{\theta}{2\pi} \oint dq = i\theta w, \quad w \in \mathbb{Z}, \quad (\text{A.3})$$

and the Euclidean weight is $e^{-S_E + i\theta w}$. Forgetting this i (i.e. writing $e^{-\theta w}$) is a classic error; the same rule governs $\theta F \tilde{F}$ in 4d and Chern–Simons terms in 3d.

Symmetry and CFT conventions. When the chapters reach the symmetry material (Part II) we follow the conventions of Shao’s TASI lectures [2308.00747](#); for CFT material (Part III), Simmons-Duffin [1602.07982](#). Conventions for p -form gauge fields, flux normalization, and Dirac matrices will be pinned here when first needed.